Digital Baseband Transmission

- Characteristics of Digital Baseband Signal
- Common Transmission Code Types
- Pulse Shaping
- Error Performance
- Eye Pattern
- Equalization

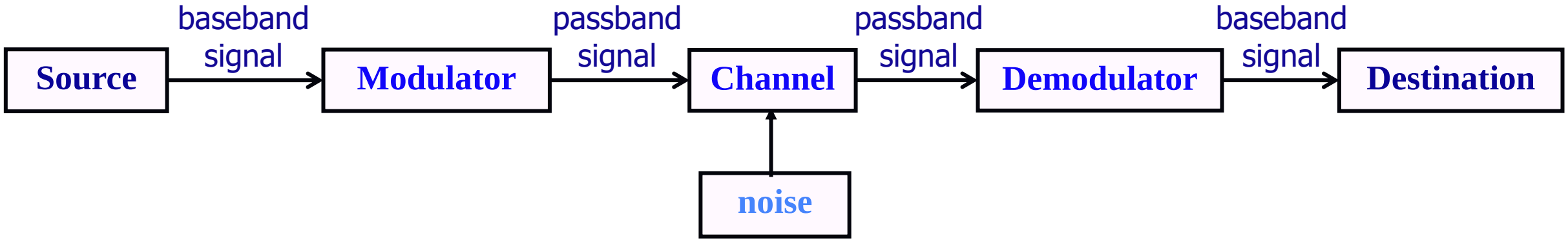
- Analog vs. Digital communication
- Baseband vs. Passband transmission
- Digital baseband vs. passband transmission

5.0 OVERVIEW OF BASIC CONCEPTS

Analog Communication System Model

- Transmitting analog signal

Analog Modulation



Analog Communication System
Modulation & Demodulation
Modeling & Model Analysis

Digital signal vs. Analog signal

Digital signal

- **Limited messages in a period of time**
- **Temporal discrete signal**
- **Limited signal： 0/1**

- Discrete symbol sequence, e.g. text message

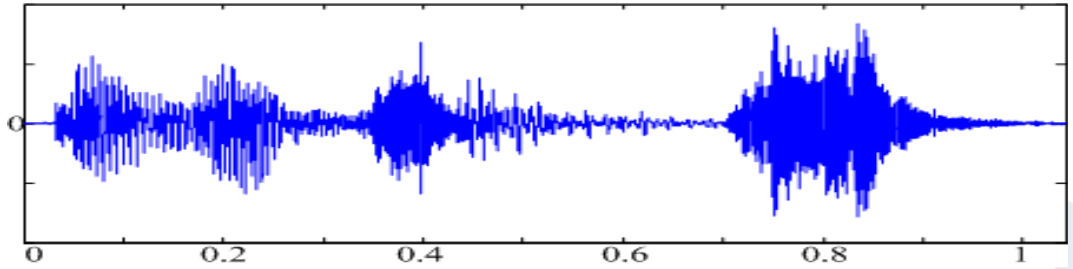
·– – ···· ·– – ···· ·– – ····

W H A T H A T H

Telegraph: The earliest digital communication method that uses electrical signals to transmit text messages

Analog signal

- **Unlimited value of messages**
- **Temporal continuous signal**
- **Messages： real value**

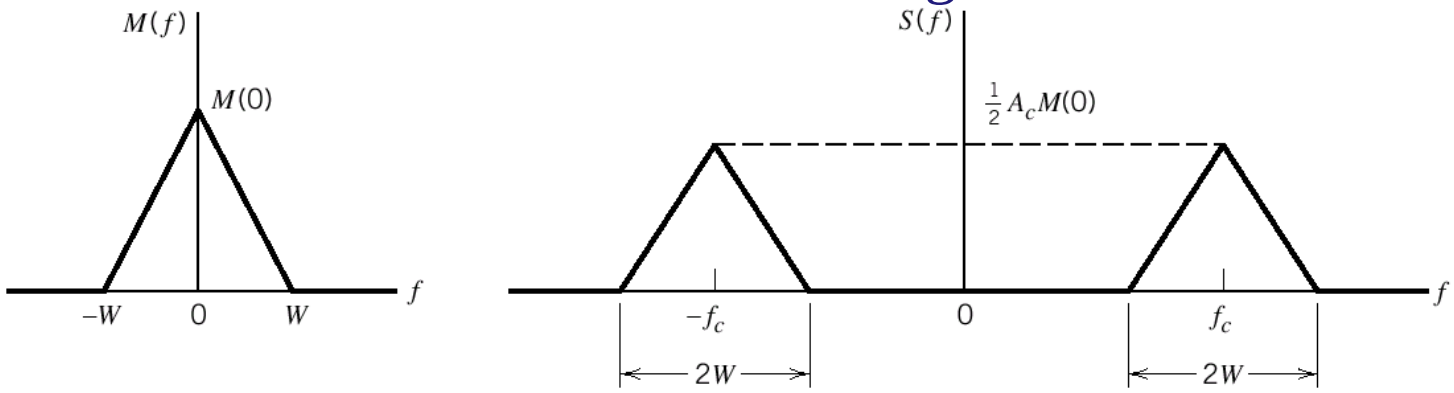


Telephone: The earliest analog communication method using electrical signals to transmit voice messages

Baseband vs. Passband transmission

- Baseband signal
  - Signals with a spectrum around zero frequency
- Passband signal
  - Signals with a spectrum around high frequencies

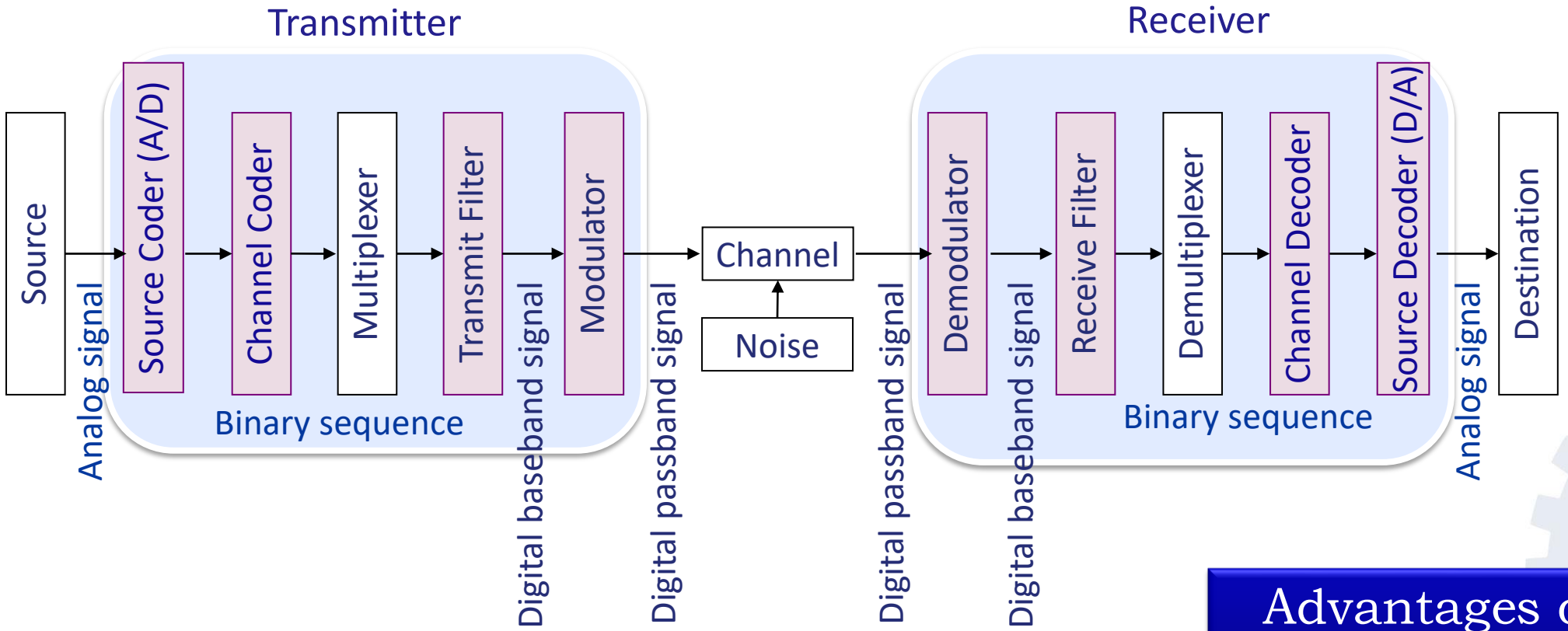
- Baseband transmission
  - transmitting baseband signal
  - wired (coaxial able, twisted pair)
  - short range
- Passband transmission
  - transmitting passband signal which is modulated signal
  - wireless or optical
  - long distance



(a) (b)

Digital Communication System Model

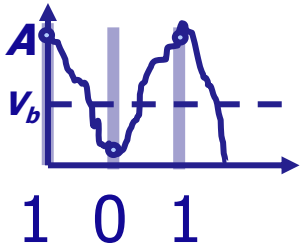
- Transmitting digital signal

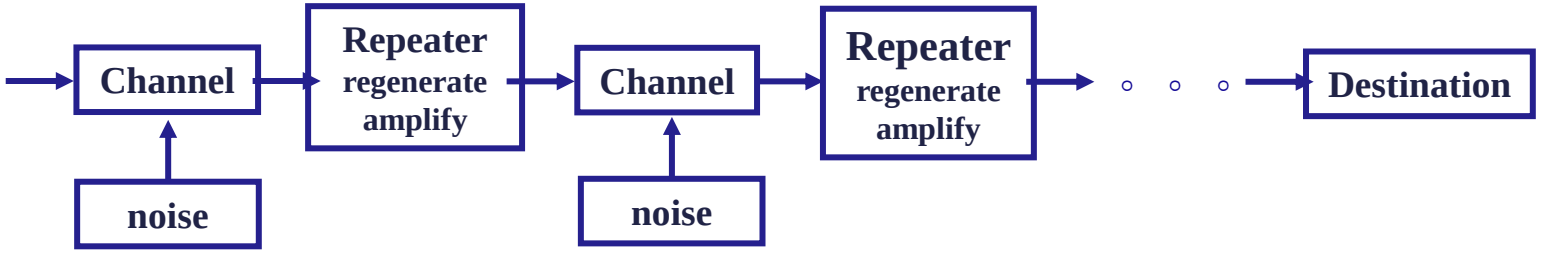


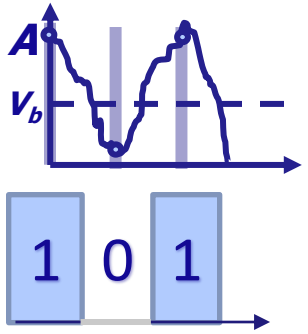
Advantages of Digital Communication?

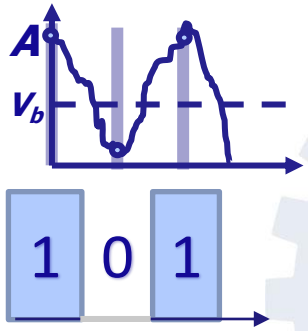
Advantages of Digital Communication

- Root causes: limited message types
  - Only 0/1 decision at receiver: classification
- High anti-noise performance
  - No noise accumulation
    - Repeater: regenerate & amplify
    - For long distance transmission
- Universal & flexible
  - 0 &1 represent data of any source
  - Easy to reuse, encrypt, encode…
  - Easy to combine with other digital technologies









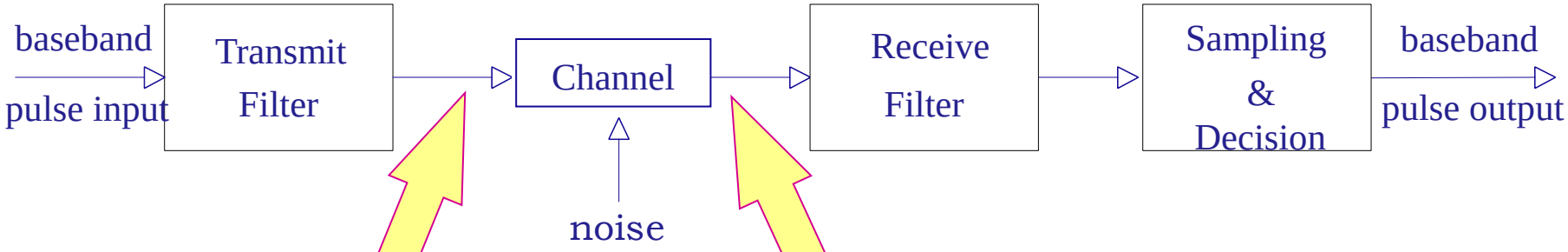
Disadvantages?

- Low bandwidth efficiency
- Synchronization need more complicated equipments

Digital baseband vs. passband transmission

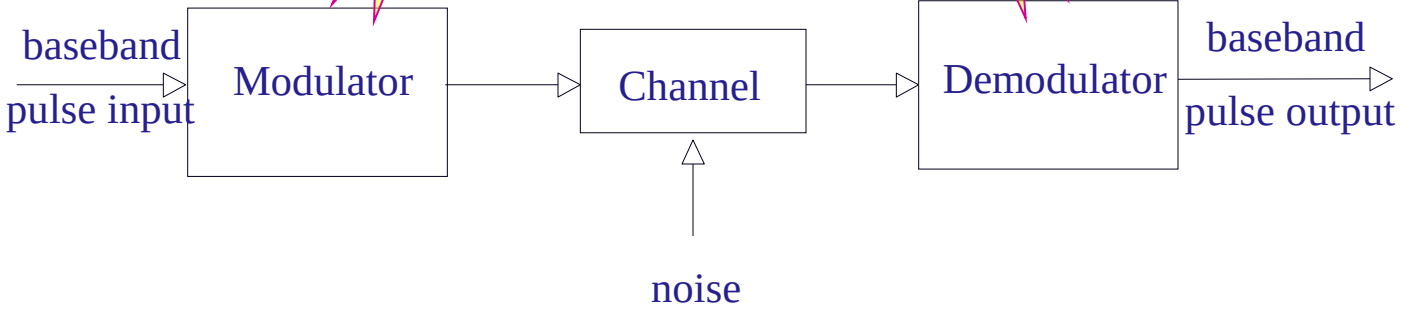
Digital baseband transmission model

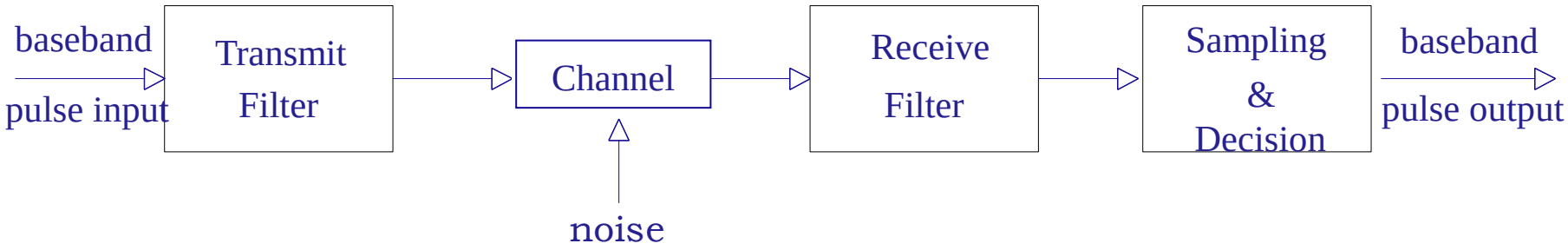
Message <==> Digital baseband Signal



Digital passband transmission model

Digital baseband Signal <==> Digital passband Signal





Digital baseband Signal: What Characteristics ?

- Transmission code types, power spectrum

Digital baseband transmission System: How to correctly transmit?

- ISI, Nyquist criterion
- Equivalent lowpass system, Partial response system

Anti-noise performance: How to analyze? How to improve?

- Reliability analysis: Symbol error rate, Eye pattern
- Transmission compensation for nonideal channel: Equalization

LEARNING GUIDE FOR THIS CHAPTER

- Basic Code Types of Digital Baseband Signal
- Power Spectrum of Digital Baseband Signal

5.1 CHARACTERISTICS OF DIGITAL BASEBAND SIGNAL

5.1.1 BASIC CODE TYPES OF DIGITAL BASEBAND SIGNAL



Binary Sequence & Binary PAM Signal

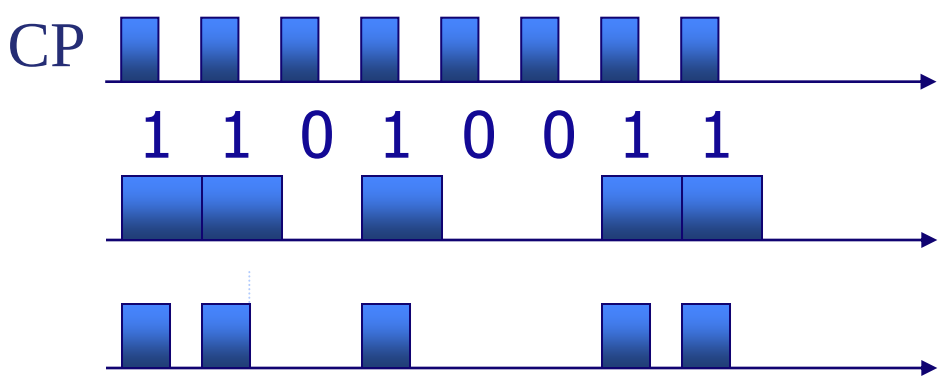
- Binary sequence
  - Binary representation of data: 0, 1 valued bit sequence
- Binary pulse amplitude modulation (2-PAM)
  - Carrier: Pulse sequence
  - Amplitude: two kinds of voltages responding to “0” & “1” separately
- Categories of pulse waveform
  - Return-to-zero (RZ) / Non-return-to-zero (NRZ)
  - Unipolar / Polar
  - absolute / relative

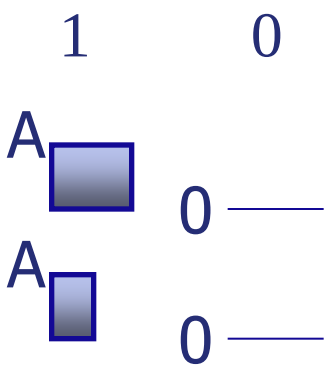


Return-to-zero / Non-return-to-zero

Clock Pulse

Bit Sequence



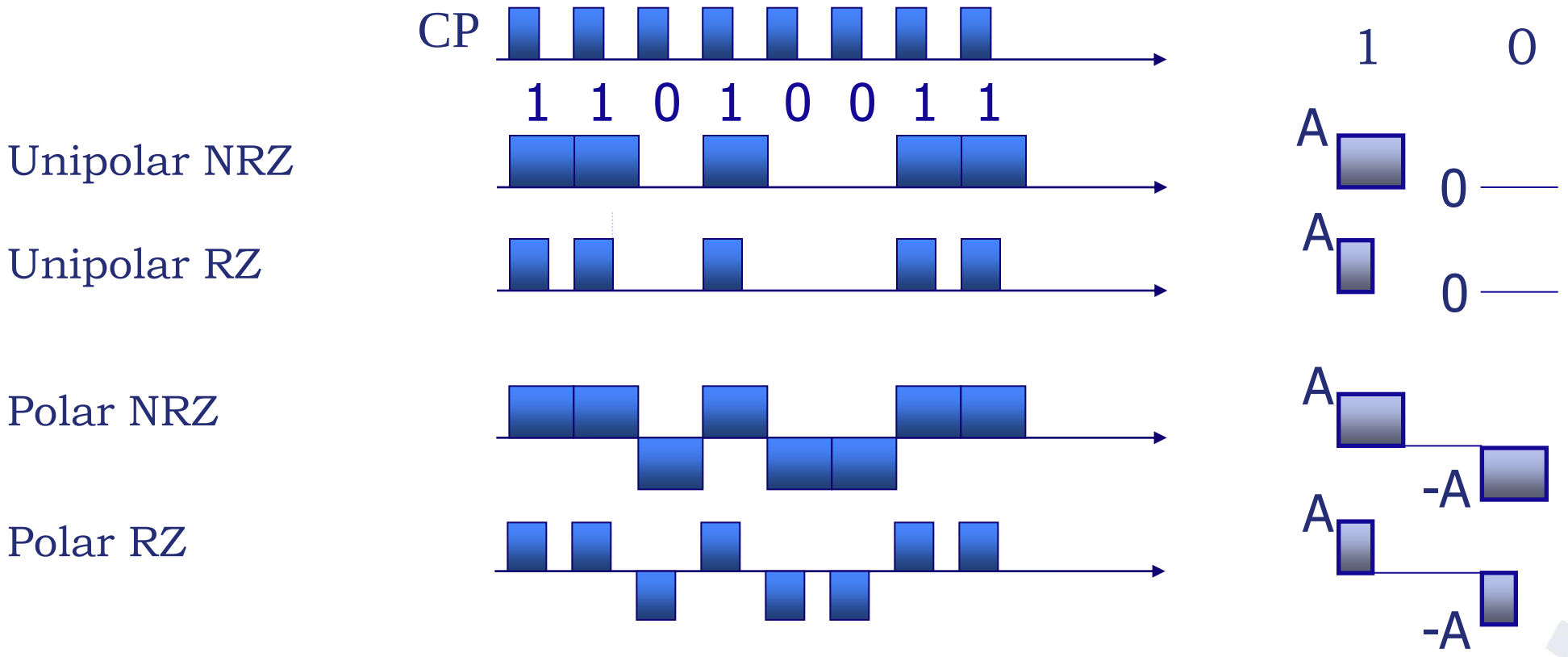


T： Code Period

τ： Pulse Width

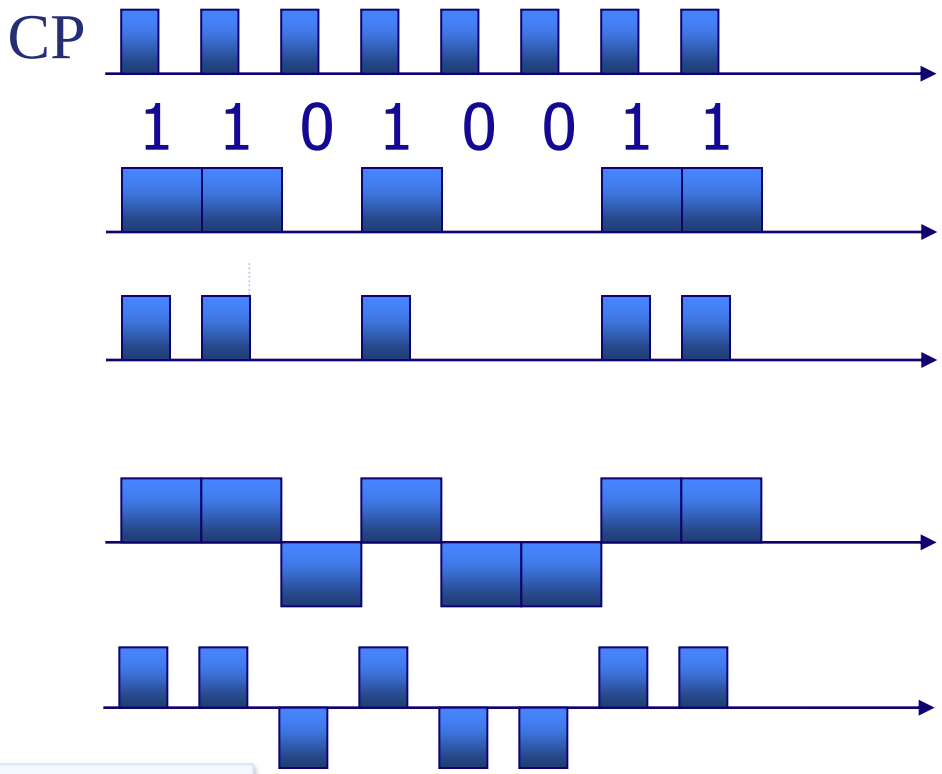
τ/T： Duty Ratio

Unipolar / Polar



Absolute code / Relative code

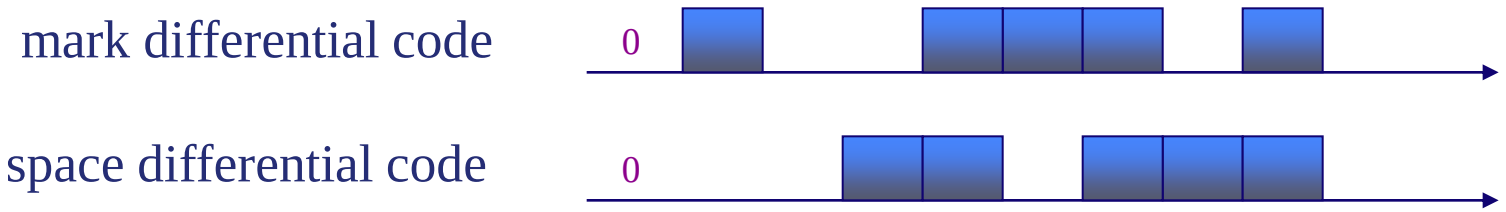
Absolute code



voltage ←→ 0,1

consideration:
write out the logic expression of the following two kinds of differential codes

Relative code: Differential code



voltage difference ←→0,1

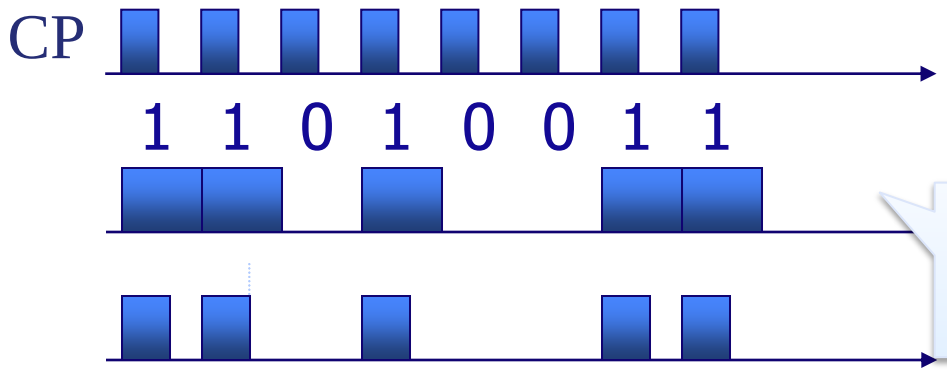
1~ change

0 ~ change

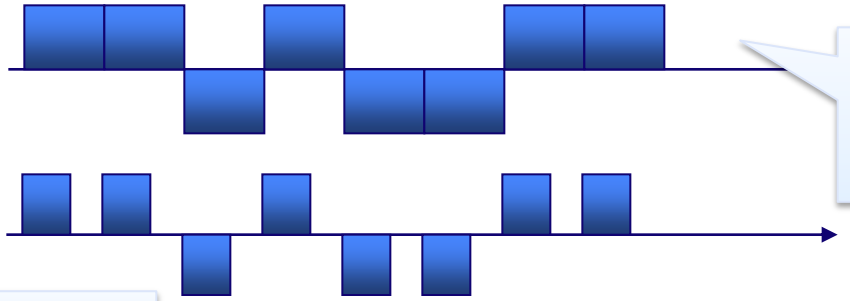
1： mark
0： space

Application of Basic Code Types

Absolute code

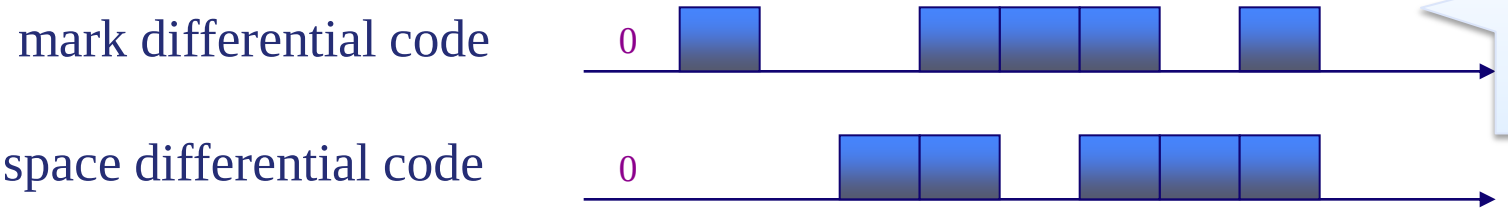


short range transmission: inside devices



RS-232 interface / V series interface

Relative code: Differential code



code converter in phase modulation system

COMMON CODE TYPE SUMMARY

RZ/NRZ

Unipolar/polar

absolute/relative

5.1.2 POWER SPECTRUM OF DIGITAL BASEBAND SIGNAL



Outline

- Characteristics of digital baseband signal — frequency domain
  - have DC component or not?
  - extract synchronization or not?
  - effective bandwidth ?
- How to analyze?
  - representation in time domain: Digital random signal
    - statistical characteristics
  - Fourier transform: Power Spectrum

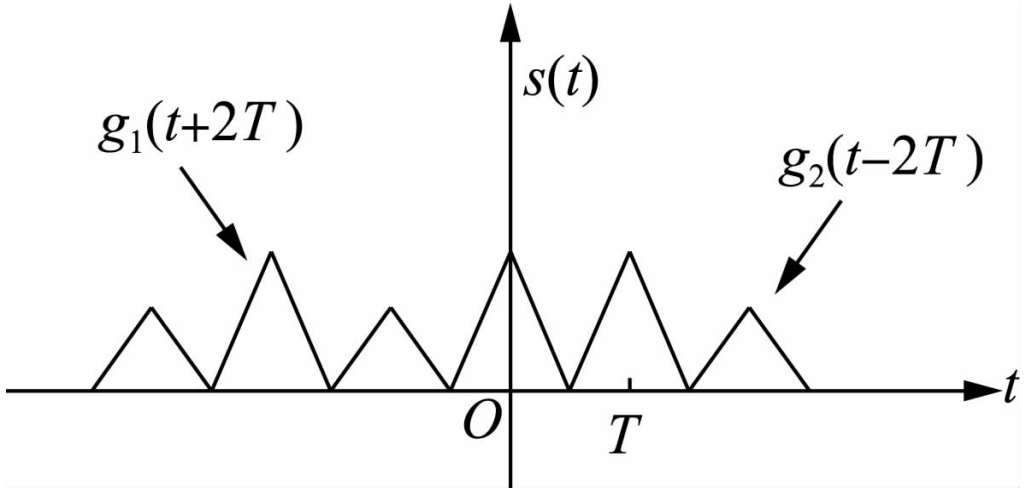


Waveform of Digital Baseband Signal

- Binary random pulse sequence
  - $g_1(t)$ and $g_2(t)$: waveform of "0" and "1"

$$s_n(t)=\begin{cases} g_1(t-nT_S) & ，\ 以概率\ P\ 出现 \\ g_2(t-nT_S) & ，\ 以概率(1-P)出现 \end{cases}$$

T_s — Code Period



$$s(t)=\sum_{n=-\infty}^{\infty}s_n(t)$$

Random Pulse Sequence in Time Domain

- baseband signal $s(t) = \sum_{n=-\infty}^{\infty} s_n(t)$

- consists of two part: stationary wave and alternative wave

$$s(t) = v(t) + u(t) \qquad P_s(f) = P_v(f) + P_u(f)$$

periodic, stationary random, alternating period, stationary random, alternating

- stationary wave

$$v(t) = \sum_{n=-\infty}^{\infty} [Pg_1(t - nT_s) + (1-P)g_2(t - nT_s)] = \sum_{n=-\infty}^{\infty} v_n(t)$$

statistical average of random sequence s(t), depending on the probability of $g_1(t)$ and $g_2(t)$

- alternating wave

$$u(t) = s(t) - v(t)$$

Power Spectrum of Digital Baseband Signal

$$P_s(f)=\sum_{m=-\infty}^{+\infty}\left|f_s[pG_1(mfB)+(1-p)G_2(mf_s)]\right|^2\delta(f-mf_s)$$

discrete spectrum

$$+f_s p(1-p)\left|G_1(f)-G_2(f)\right|^2$$

continuous spectrum

$$g_1(t)\leftrightarrow G_1(f)$$
$$g_2(t)\leftrightarrow G_2(f)$$
$$f_s=1/T_s$$

***p* is the prior probability** of transmitting **symbol 1**

- Related factors
  - $G_1(f)$ and $G_2(f)$
  - prior probability ~ p
  - code rate ~ f_s

- Continuous spectrum must exist
- Discrete spectrum may not exist

Discussion: derivation process of the power spectrum

Unipolar Code

- Let $g_1(t)=0$, $g_2(t)=g(t)$, then the power spectrum of baseband signal is

$$P_s(f)=\sum_{m=-\infty}^{+\infty}\left|f_s(1-p)G(mf_s)\right|^2\delta(f-mf_s)+f_sP(1-p)\left|G(f)\right|^2$$

- $g(t)$ ~ rectangular-shaped pulse
  - duty ratio ~ τ/T

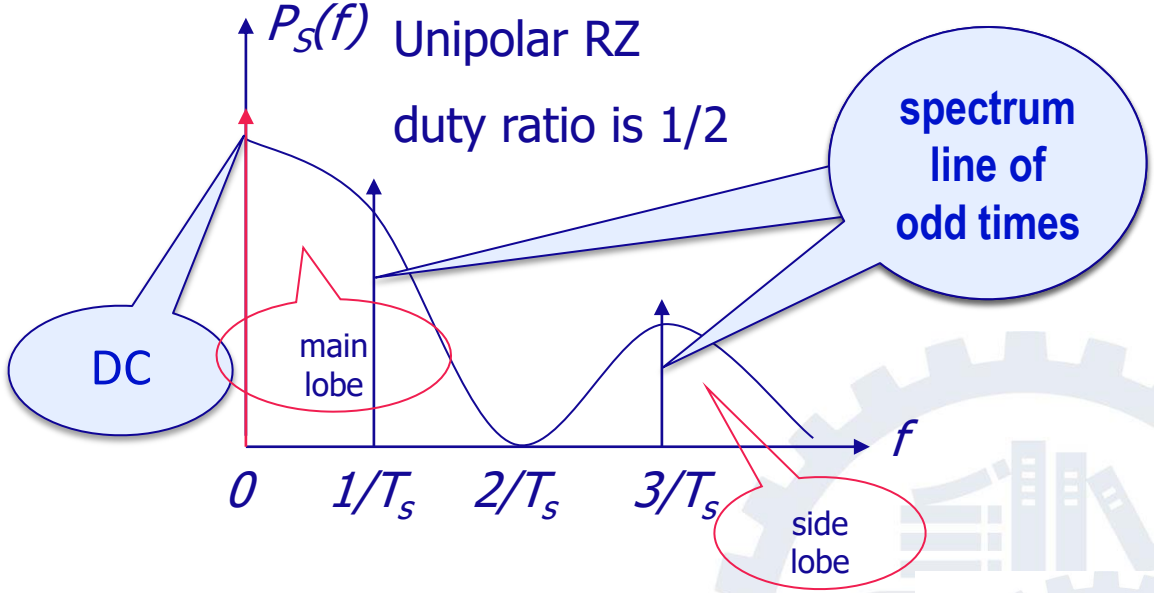
$$g(t)=\begin{cases}1, |t|\le\dfrac{\tau}{2}\\ 0, |t|>\dfrac{\tau}{2}\end{cases}$$

  - spectrum of single pulse

$$G(f)=\tau Sa(\pi f\tau)$$

  - if p =1/2, then the spectrum is

$$P_s(f)=\frac{f_s^2\tau^2}{4}\sum_{m=-\infty}^{+\infty}Sa^2(\pi mf_s\tau)\delta(f-mf_s)+\frac{f_s\tau^2}{4}Sa^2(\pi f\tau)$$



Power Spectrum of Basic Code Types

$$P_s(f)=\sum_{m=-\infty}^{+\infty}\left|f_s[pG_1(mf_s)+(1-p)G_2(mf_s)]\right|^2\delta(f-mf_s)$$
$$+f_s p(1-p)\left|G_1(f)-G_2(f)\right|^2$$

- Unipolar NRZ

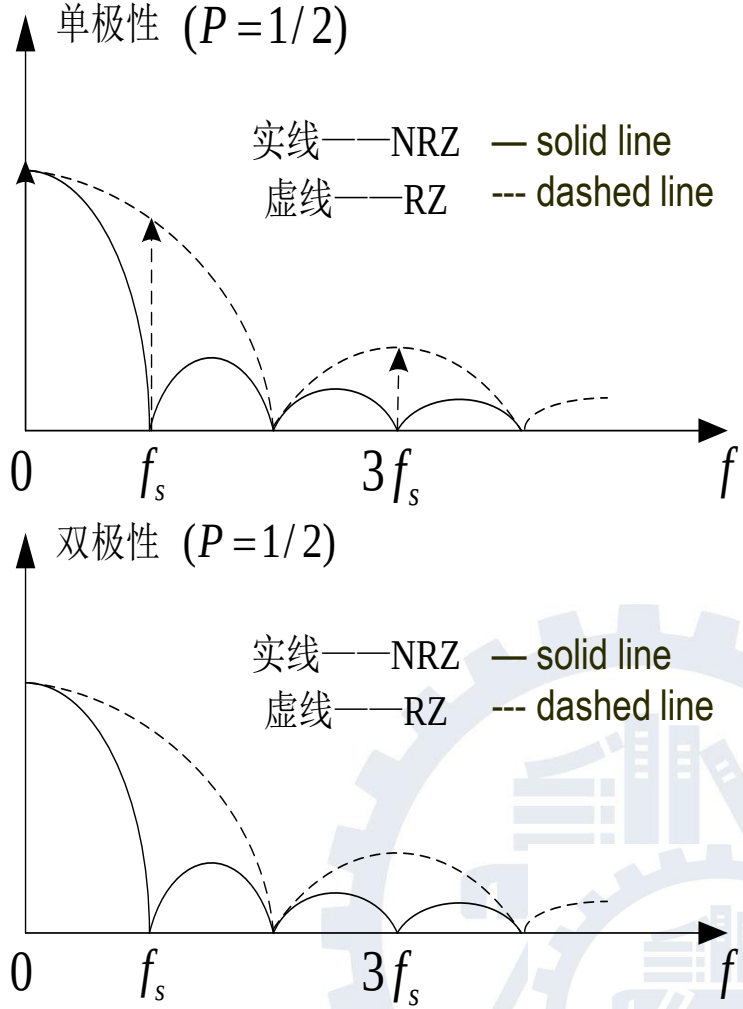
$$P_s(f)=\frac{T_s}{4}Sa^2(\pi fT_s)+\frac{1}{4}\delta(f)$$

impulse at 0 freq.

- Unipolar RZ
  - duty ratio is 1/2

$$P_s(f)=\frac{T_s}{16}Sa^2(\pi fT_s)+\frac{1}{16}\sum_{m=-\infty}^{\infty}Sa^2(\frac{m\pi}{2})\delta(f-mf_s)$$

impulse at 0 and odd times



- Polar NRZ

$$P_s(f)=T_s\cdot Sa^2(\pi fT_s)$$

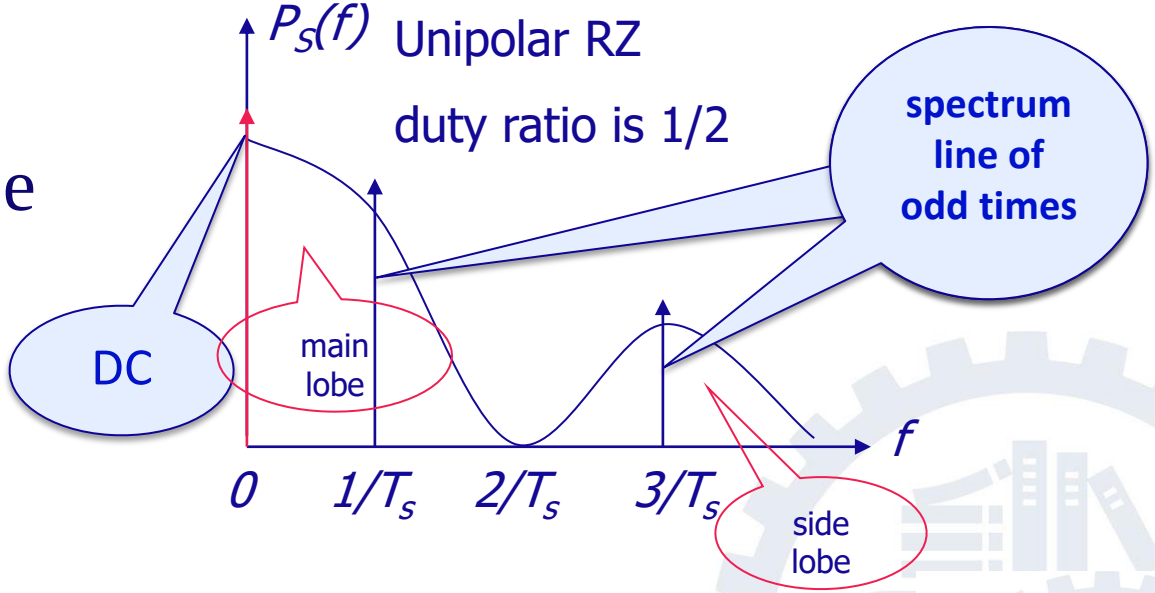
- Polar RZ
  - duty ratio is 1/2

$$P_s(f)=\frac{T_s}{4}\cdot Sa^2(\frac{\pi fT_s}{2})$$

no impulse exists

Characteristics of Power Spectrum

- Polar Code:
  - if p =1/2, then neither DC component nor clock component exists
- Unipolar code has DC component
  - Unipolar RZ has rich clock component
- effective bandwidth: width of main lobe
  - NRZ: $B = 1/T_s = f_s$ Hz
  - RZ: $B = 1/\tau$
  - lower duty ratio, wider frequency band



POWER SPECTRUM OF BASEBAND SIGNAL

- Power spectrum depends on pulse shaping and prior prob, not phase
- Continuous spectrum always exists, not necessary for discrete spectrum
- BW of main lobe depends on the width of pulse shaping

- Principles of Choosing Transmission Code Types
- Common Transmission Code Types
- HDB_3 Code

5.2 COMMON TRANSMISSION CODE TYPES

5.2.1 PRINCIPLES OF CHOOSING TRANSMISSION CODE TYPES

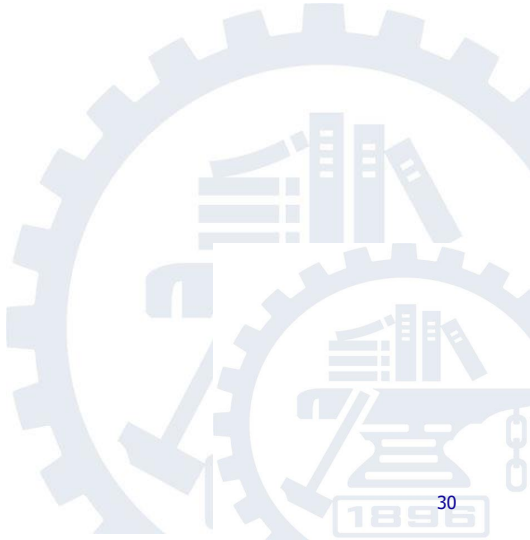


Principles for Choosing Transmission Code Types

Requirements for Transmission	Characteristics of Transmission Code Types
• Synchronization	• easy to get clock
• Limited low freq.	• no DC, little low frequency
• High efficiency	• convergent spectrum
• High reliability	• low bit error rate
• High flexibility	• known at source
• Low cost	• simple coding & decoding

Are the basic code types suitable for transmission of long distance?

5.2.2 COMMON TRANSMISSION CODE TYPES



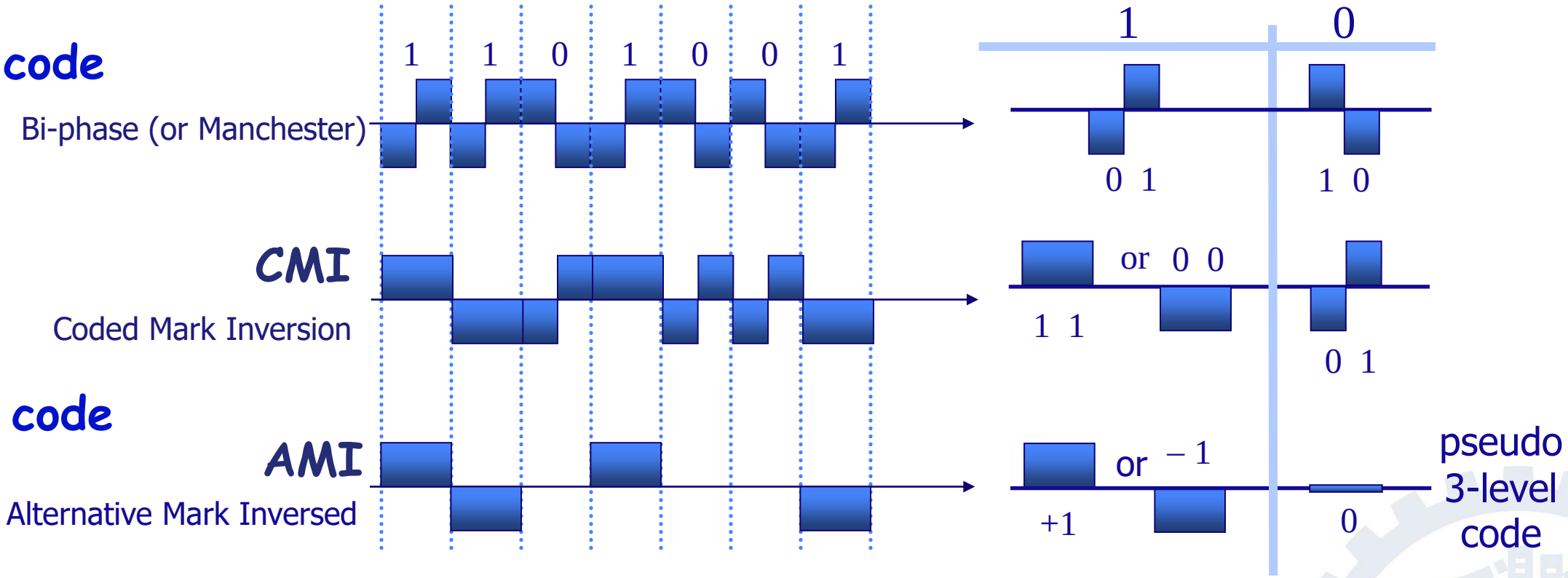
Common Transmission Code Types

- Two-level code
  - nBmB
- Pseudo multi-level code
  - kBnT
- Multi-level code



1B2B, 1B1T

1B2B code



1B1T code

Manchester code:
no DC; easy clocking;
double width;
Ethernet line code

CMI:
no DC; easy clocking;
double width; error detection
PCM E4 line code

AMI:
no DC; error detection;
easy clocking while RZ
PCM T1 line code

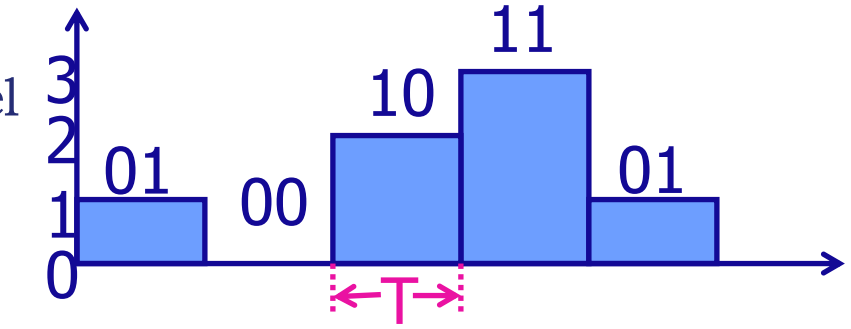
nBmB、nBmT

- Decrease the cost of band expansion

- nBmB: take n bits as a group, convert to m bits for transmission
  - Optical digital transmission system: usually choose m＝n＋l, e.g. 1B2B, 2B3B and 5B6B, etc.
- nBmT: take n bits as a group, convert to m Ternary symbols for transmission
  - 1B1T：AMI
  - 8B6T: 100Mbps Ethernet

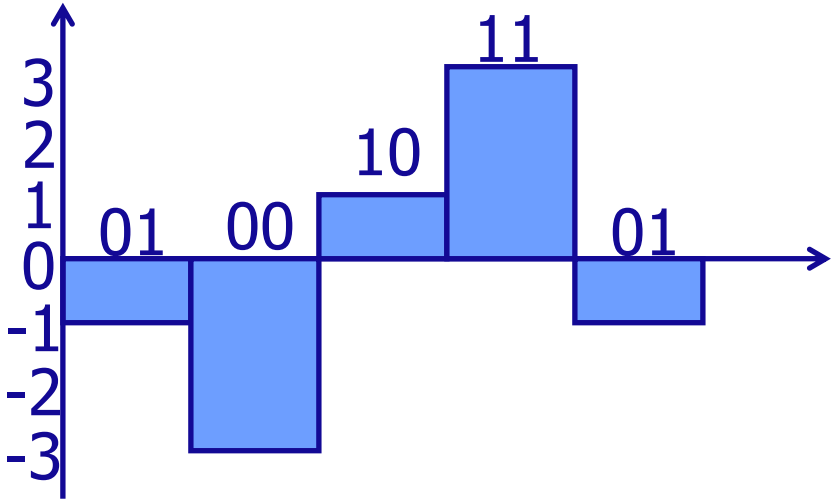
Multilevel Code

Multilevel



code	00	01	10	11
voltage	0	1	2	3

2B1Q



code	00	01	10	11
voltage	-3	-1	1	3

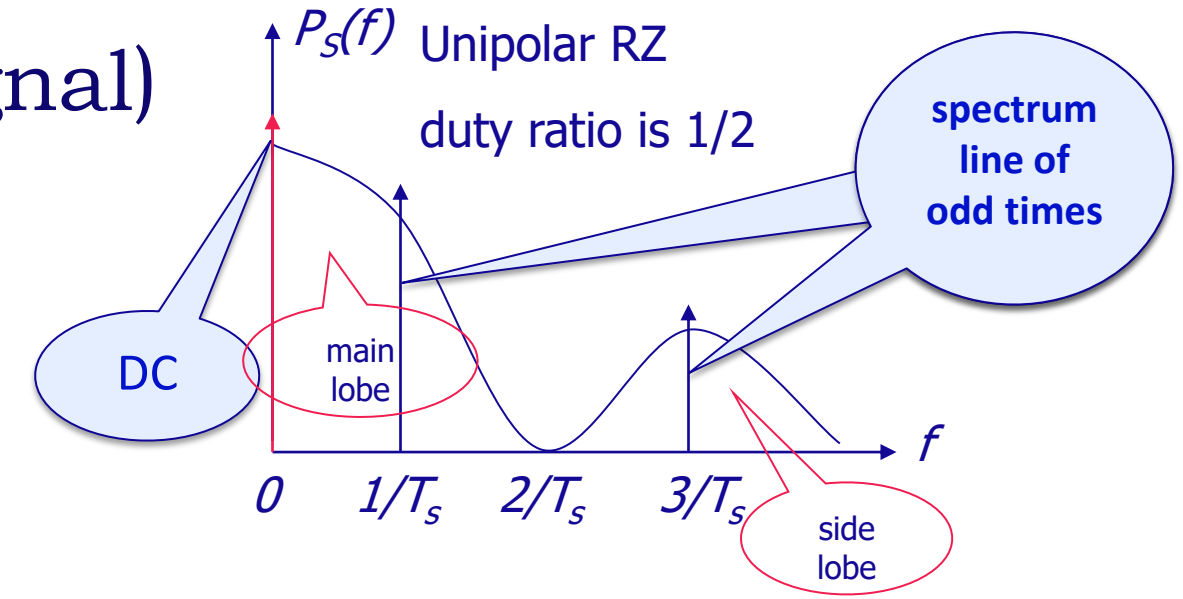
- $M=2^k$, $k=2$

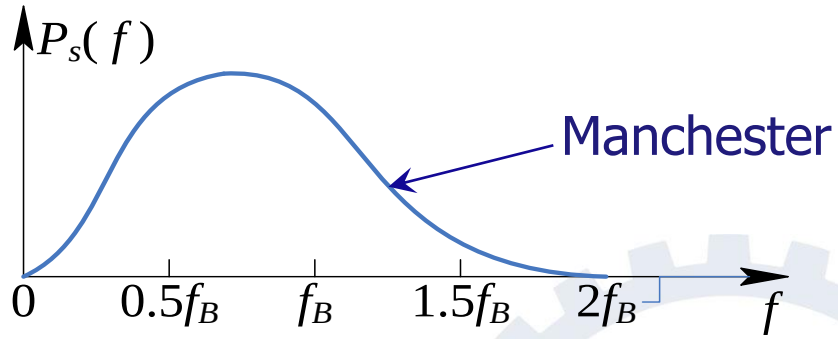
for given bit rate, the band width is B/k

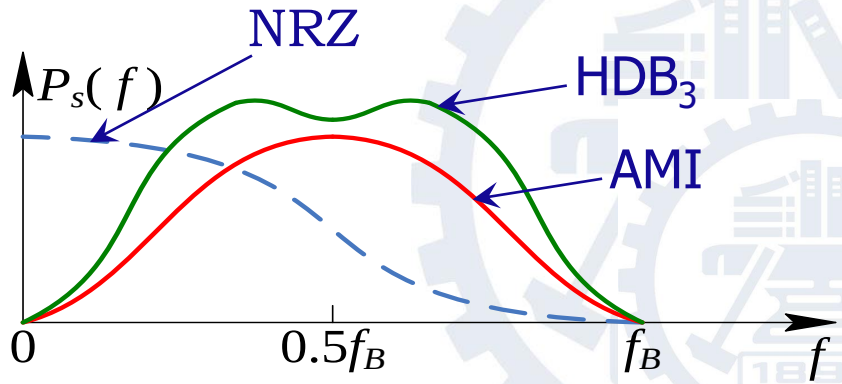
Effective Bandwidth (M-ary Signal)

- Most of the energy locate in the main lobe of power spectrum, so, the effective bandwidth is the width of the main lobe
  - NRZ: $B = 1/T_s = R_B$ Hz
  - RZ: $B = 1/\tau$
- Transmission rate
  - $R_b = log_2 M \, R_B$ (bit/s)

Code Type	Effective Bandwidth
CMI, bi-phase code	$2/T_s$
NRZ, AMI, HDB_3	$1/T_s$
RZ	$1/\tau$



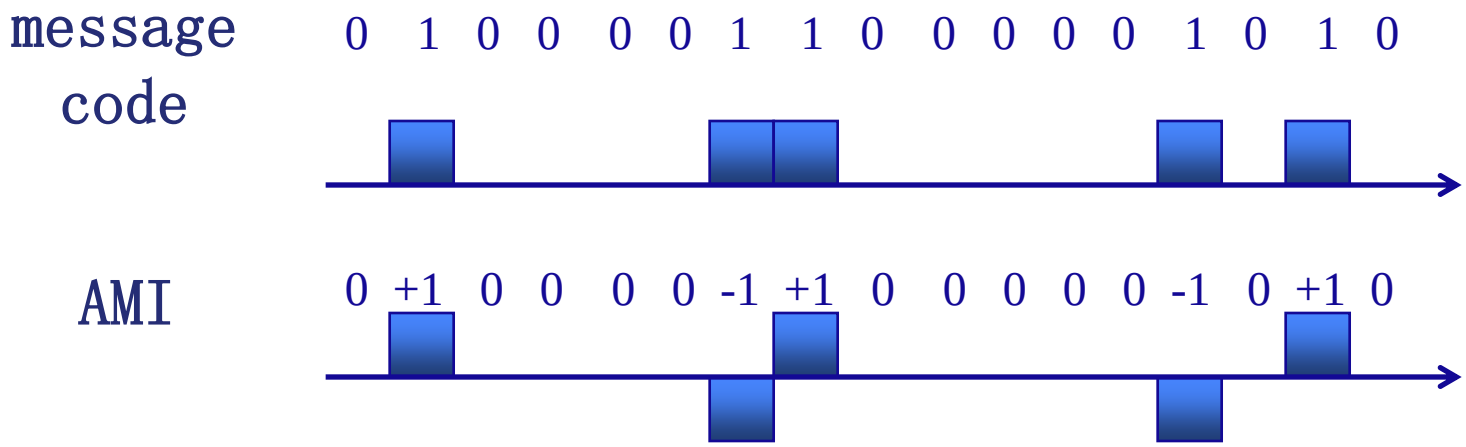




5.2.3 HDB_3 CODE



The Problem of Long 0 Strings in AMI: How to Overcome?



In case of long 0 strings, AMI is hard to extract sync signals

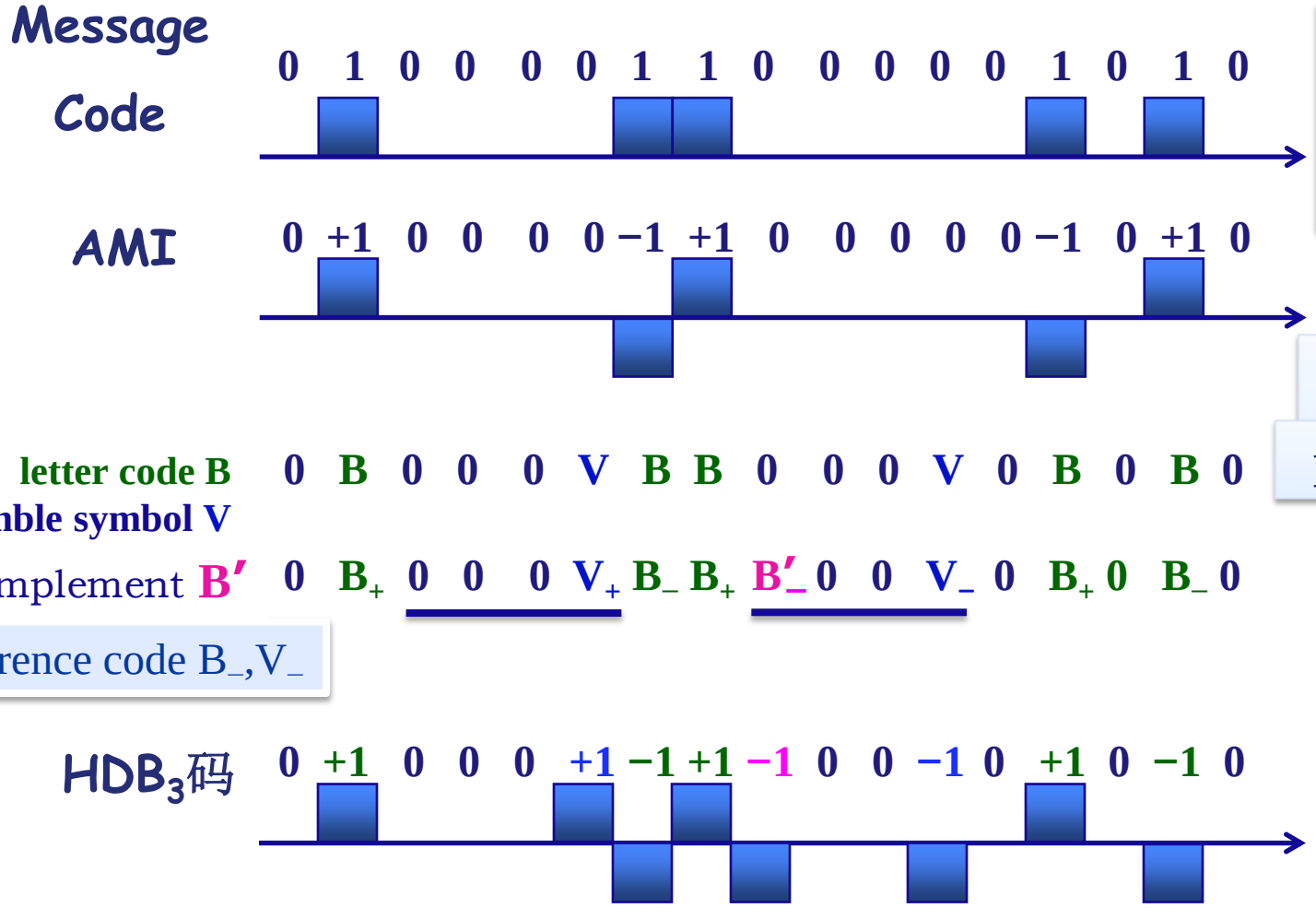
How to overcome this problem?

- HDB3 (High-Density Bipolar code of order 3)
  - Scrambling (by "1") → number of "0" strings ≤ 3
  - PCM E1 line code

How to design scrambling rule?

Encoding Procedure of HDB_3 Code

How to design scrambling rule?



function: break successive 0

condition:

scramble symbol can be recognized

maintain polarity alteration balance

the 4th "0" → V code

polarity alteration betw. B code

polarity alteration betw. V code

V code and the former B code should have the same polarity

add complement code **B′**, otherwise

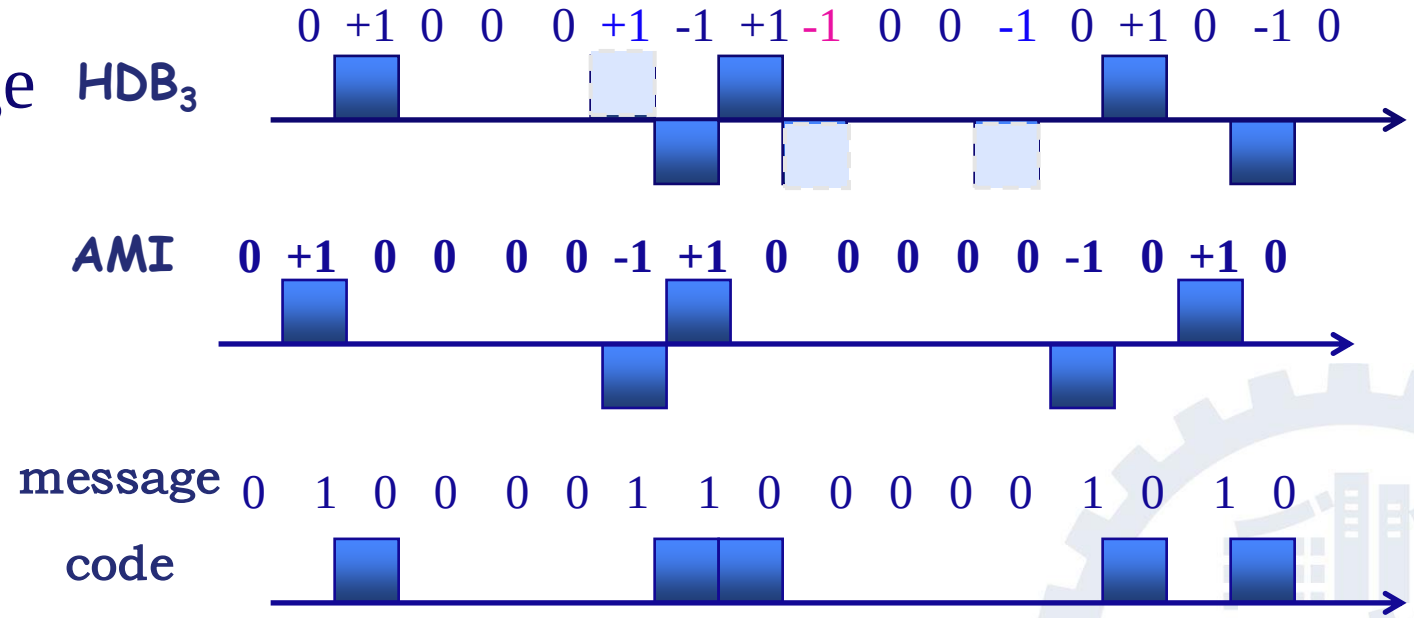
set reference code B−,V−

two kinds of scrambling segments: 000V, **B′**00V

How to descrambling ?

Decoding Procedure of HDB_3 Code

- Recognize scramble symbol
- Eliminate V、B’，then get AMI code
- Decode AMI code to message code by full-wave rectification



Characteristics of HDB_3 Code

Encoding message → AMI → 4 "0" → add destruction symbol

Decoding recognize → eliminate → AMI → message

- Advantages
  - no DC、few low freq.
  - convergent spectrum
  - easy to extract sync signals
- Disadvantages
  - encoding and decoding circuits are complex

application: Europe、China
PCM E1, E2, E3 interface line code

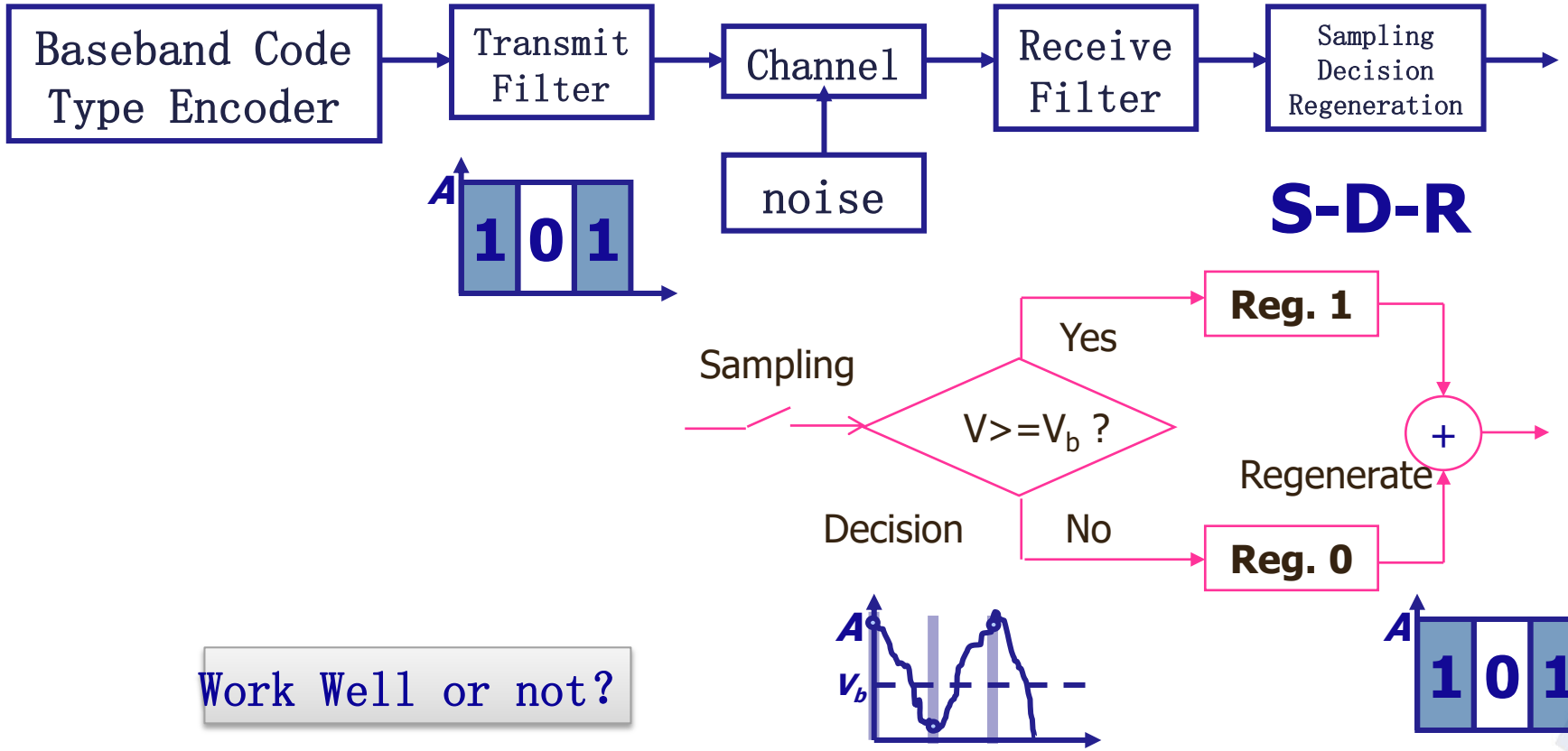
- Inter-symbol Interference
- 1st Nyquist Criterion
- Equivalent lowpass System
- 2nd Nyquist Criterion
- Partial Response System
- Optimization of Baseband Transmission

5.3 PULSE SHAPING

5.3.1 INTER-SYMBOL INTERFERENCE

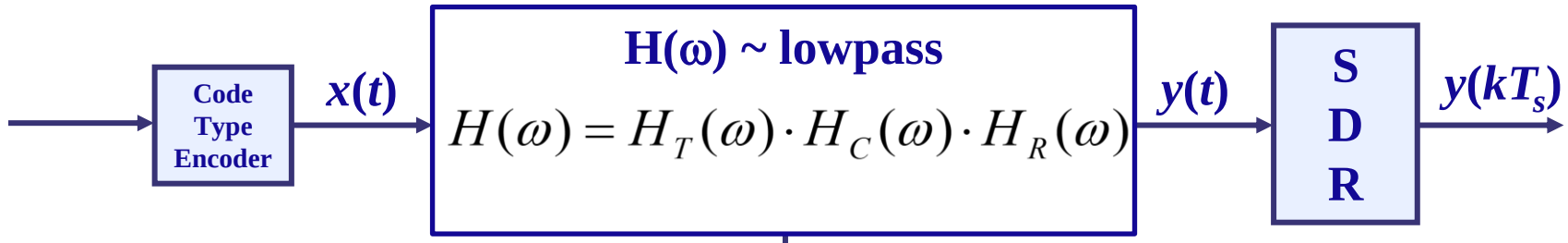


Digital Baseband Transmission System



Work Well or not?

Band-limited



- Limited in time domain

$h(t)$ is limited, $t \in (0, T_s)$

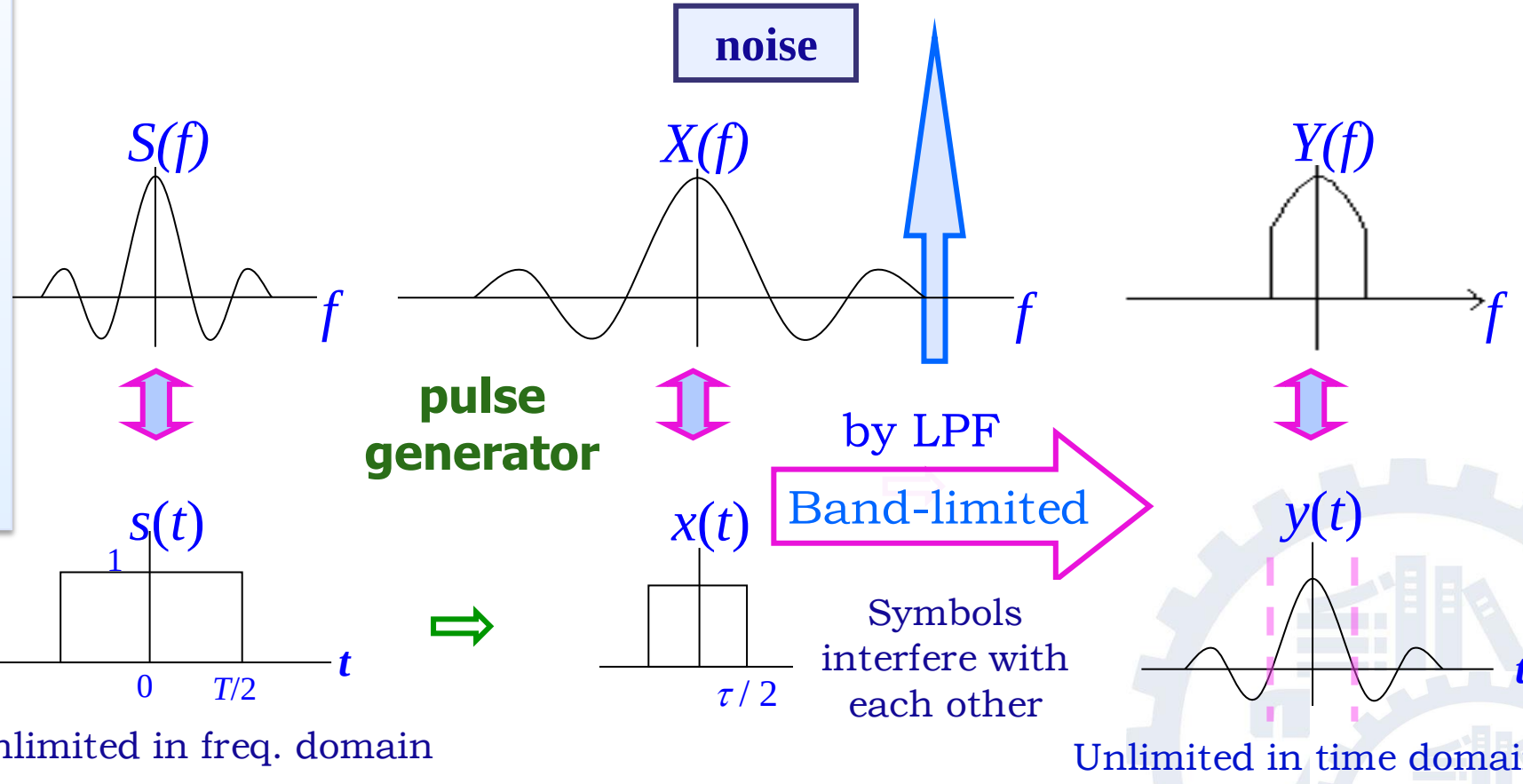
- Unlimited in freq. domain

- Limited in freq. domain

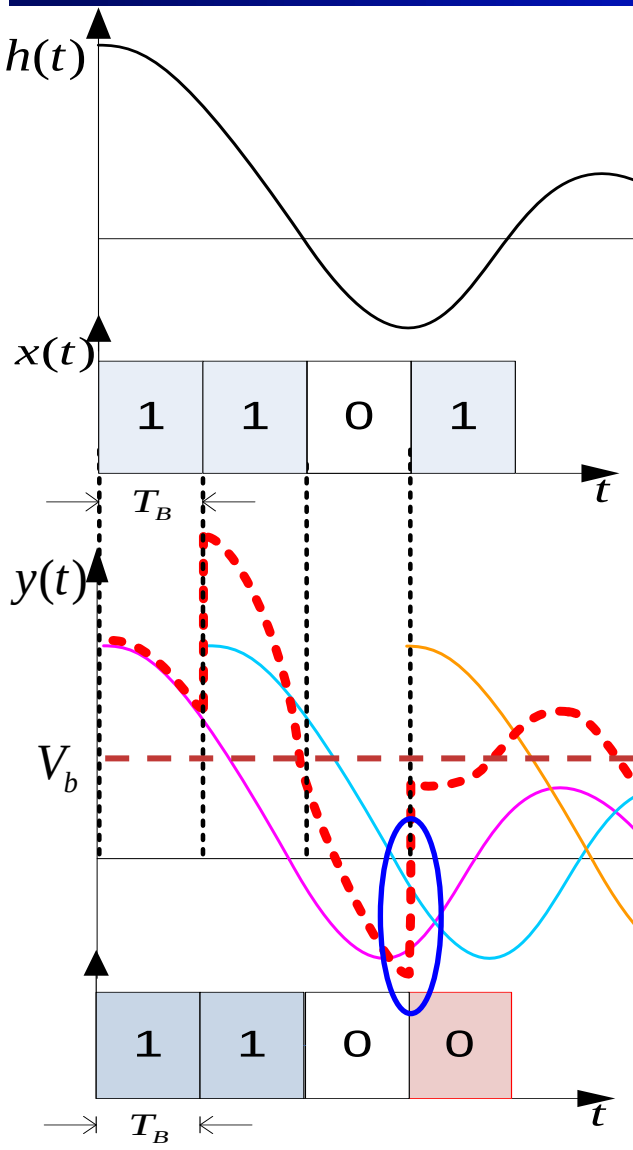
$H(f)$ is limited, $f \in (0, B)$

- Unlimited in time domain

$h(t)$ last longer than symbol period



InterSymbol Interference (ISI)



impulse response $h(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} H_T(\omega)H_C(\omega)H_R(\omega)e^{j\omega t}d\omega$

input $x(t) = \sum_{n=-\infty}^{\infty} a_n \delta(t - nT_s)$

output $y(t) = \sum_{n=-\infty}^{\infty} a_n h(t - nT_s) + n_R(t)$

kT_s moment $y(kT_s) = \sum_{n=-\infty}^{+\infty} a_n h[(k-n)T_s] + n_R(kT_s)$

$= a_k h(0) + \sum_{n\neq k} a_n h[(k-n)T_s] + n_R(kT_s)$

sample value of kth symbol

ISI

random interference

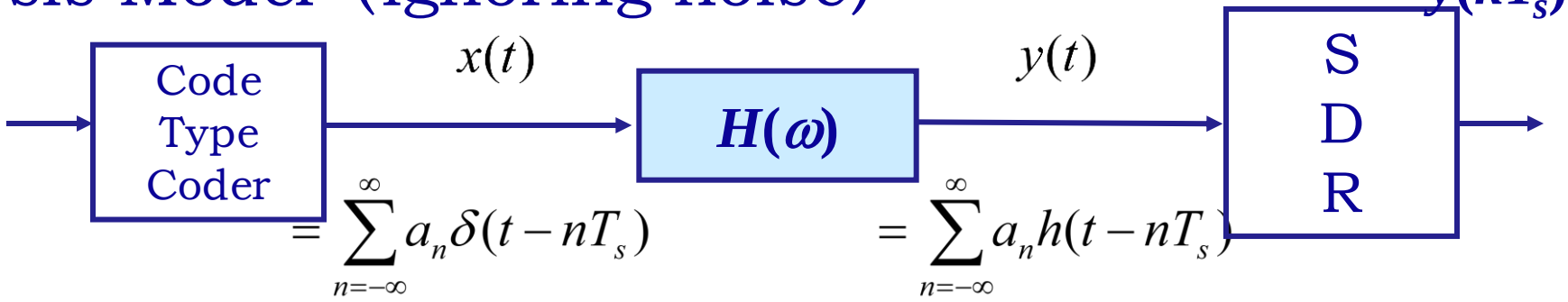
How to eliminate ISI?

5.3.2 FIRST NYQUIST CRITERION



How to Eliminate ISI ?

- Analysis Model (ignoring noise)



Sample value at kT_s moment

$$y(kT_s) = \sum_n a_n h(kT_s - nT_s)$$
$$= a_k h(0) + \sum_{n \neq k} a_n h\left[(k-n)T_s\right]$$
$$= a_k h(0)$$

have no ISI

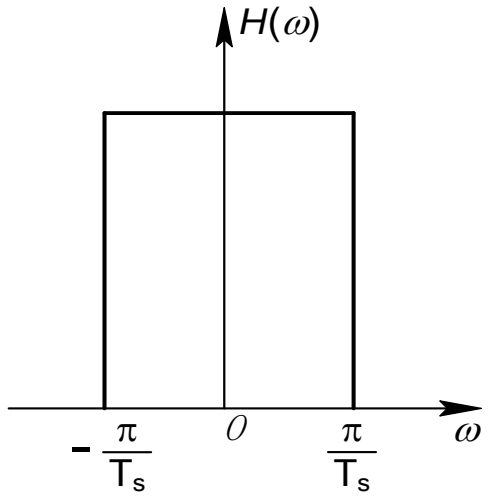
So, we have $h[(k-n)T_s] = \begin{cases} C, & k = n \\ 0, & k \neq n \end{cases}$

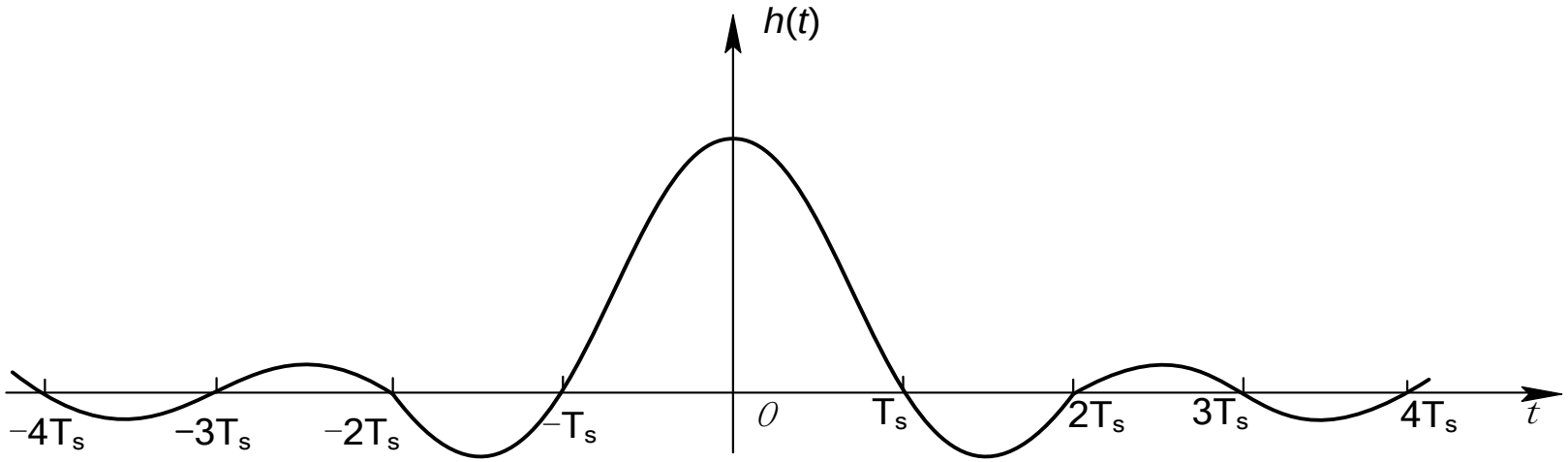
k = k – n

$$h(kT_s) = \begin{cases} C, & k = 0 \\ 0, & k \neq 0 \end{cases}$$

no ISI condition
in time domain

No ISI Condition (Time Domain)





Basic idea: the samples of system impulse response have some value for the current symbol, while exactly equal to 0 for other symbols

$$H(\omega) = rect(\omega/W) \quad W = 2\pi/T_s$$

$$h(t) = \frac{W}{2\pi} Sa\left(\frac{Wt}{2}\right) = f_s Sa(\pi f_s t)$$

h(t)?
T_s ?

$$h(kT_s) = \begin{cases} C, & k = 0 \\ 0, & k \neq 0 \end{cases}$$

No ISI condition (time domain)

1st Nyquist Criterion

Nyquist pulse-shaping criterion

System impulse response $h(t)$ satisfies $h(kT_s) = \begin{cases} C, & k = 0 \\ 0, & k \neq 0 \end{cases}$

No ISI condition (time domain)

At sampling points, No ISI

←→

System transfer function $H(\omega)$ satisfies

$$\sum_{n=-\infty}^{\infty} H(f - \frac{n}{T_s}) = C \qquad |f| \leq \frac{1}{2T_S}$$

No ISI condition (freq. domain)

equivalent lowpass

Deriving Frequency Conditions

- Inverse Fourier of impulse response
- $t = kT_s$

$$h(t) = \int_{-\infty}^{\infty} H(f) e^{j2\pi ft} df \qquad h(kT_s) = \begin{cases} 1, & k = 0 \\ 0, & k \neq 0 \end{cases}$$

1st Nyquist criterion condition (time domain)

$$h(kT_S) = \int_{-\infty}^{\infty} H(f) e^{j2\pi fkT_S} df$$

- Piecewise $(1/T_s)$ integral & summation
- Reverse the order of integral & summation

$$= \sum_n \int_{(2n-1)/2T_S}^{(2n+1)/2T_S} H(f) e^{j2\pi fkT_S} df$$

$$= \int_{-1/2T_S}^{1/2T_S} \left[\sum_i H(f + \frac{n}{T_S}) \right] e^{j2\pi fkT_S} df$$

period $1/T_s = R_B$

- Fourier series of periodic signal

$$\frac{1}{T_S} \sum_n H(f + \frac{n}{T_S}) = \sum_k h(kT_S) e^{-j2\pi fkT_S}$$

Fourier series of periodic signal

$$F(f) = \sum_n f_n e^{-jn2\pi fT_S}$$

coefficient of Fourier series

$$f_n = T_S \int_{-1/2T_S}^{1/2T_S} F(f) e^{jn2\pi T_S} df$$

Deriving Frequency Conditions

$$\frac{1}{T_S}\sum_n H(f+\frac{n}{T_S}) = \sum_k h(kT_S)e^{-j2\pi fkT_S}$$

$$h(kT_s) = \begin{cases} 1, & k=0 \\ 0, & k\neq 0 \end{cases}$$

1st Nyquist criterion condition (time domain)

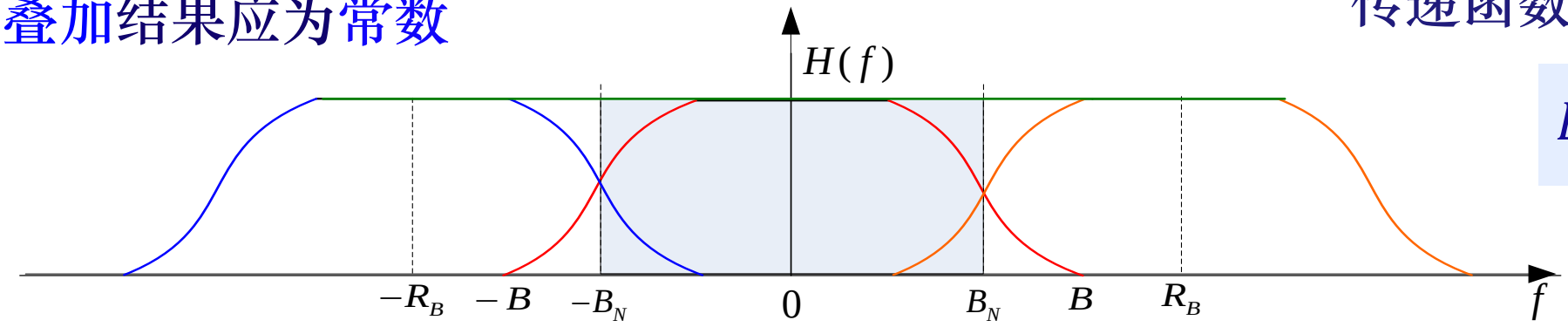
Baseband system transfer function satisfies

$$\sum_n H(f+\frac{n}{T_s}) = T_S, \qquad |f| \le \frac{1}{2T_S}$$

1st Nyquist criterion condition (freq. domain)

Equivalent lowpass

将$H(f)$在f轴上以$1/T_s$为间隔切开
沿f轴分段平移到 $(-1/2T_s, 1/2T_s)$ 区间内
叠加结果应为常数



互补对称滚降特性
传递函数关于B_N点互补对称

$$B_N = \frac{1}{2T_S} = \frac{R_B}{2}$$

5.3.3 EQUIVALENT LOWPASS SYSTEM

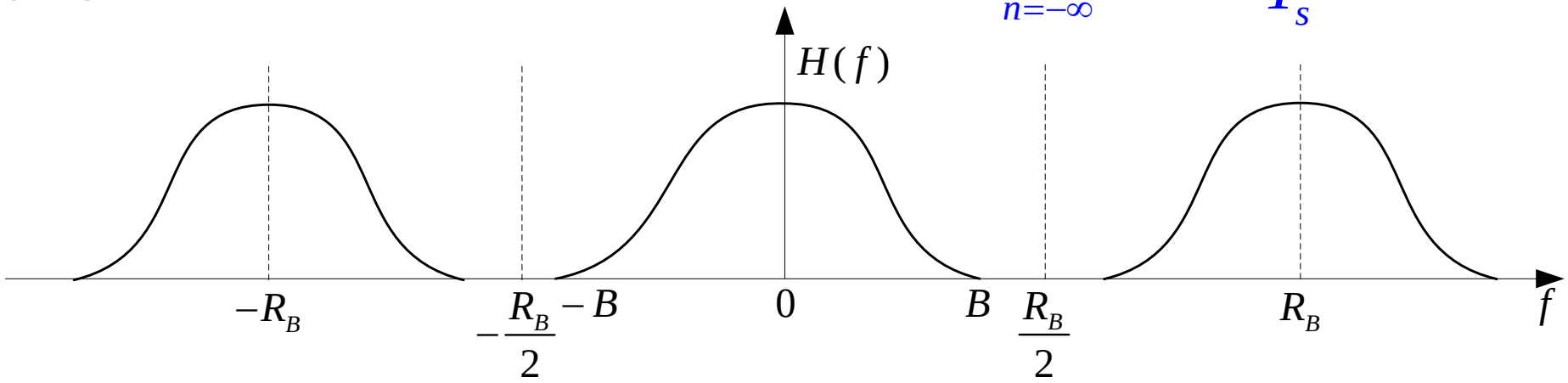


Design Equivalent Lowpass Baseband System

1st Nyquist criterion

$$\sum_{n=-\infty}^{\infty} H(f-\frac{n}{T_s})=C$$

- **1. $B<R_B/2$** no solution

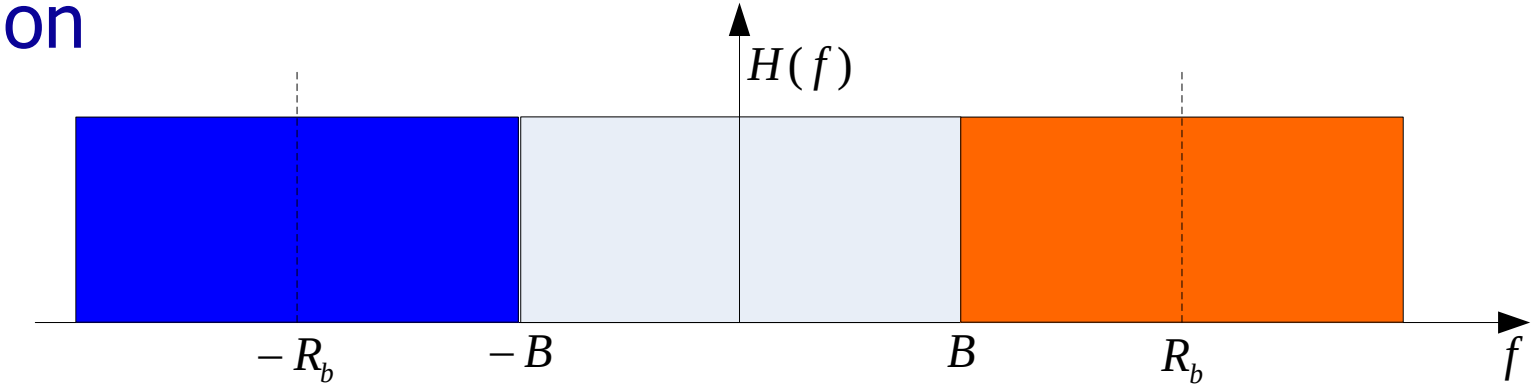


Design Equivalent Lowpass Baseband System

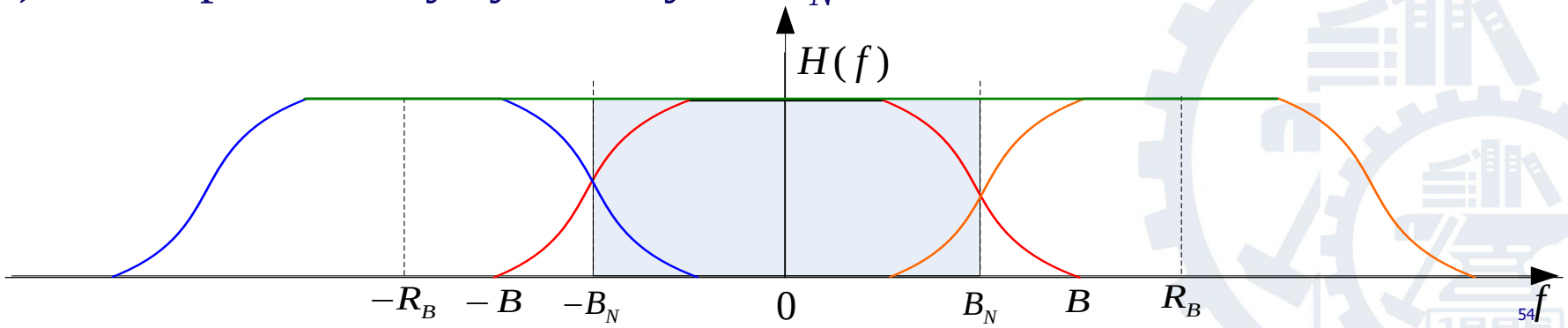
1st Nyquist criterion

$$\sum_{n=-\infty}^{\infty} H(f-\frac{n}{T_s})=C$$

- **1. $B<R_B/2$** no solution
- **2. $B=R_B/2$** unique solution
- Ideal lowpass system

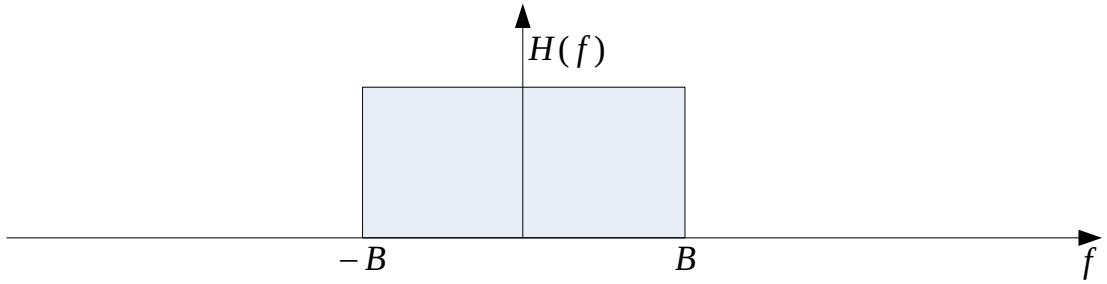


- **3. $B>R_B/2$** multi-solution
- Complementary symmetry rolloff lowpass system
  - The rolloff part of $H(f)$ is complementary symmetry at B_N



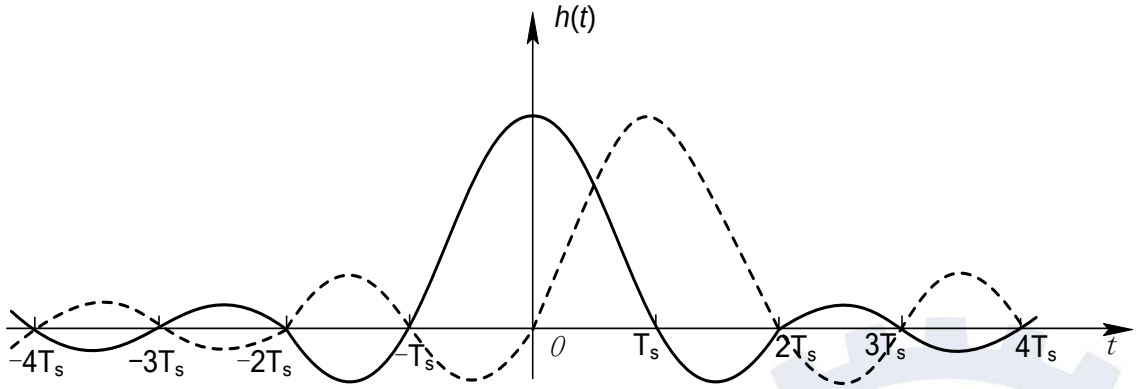
Ideal Lowpass System

- Absolute rating of Nyquist baseband transmission system
  - $B_N=R_B/2$ (Nyquist bandwidth)
  - $R_B=2B_N$ (Nyquist transmission rate)
  - Ideal bandwidth efficiency: $\eta=R_b/B=2\text{bit/s}\cdot\text{Hz}$



$$H(\omega)=T_s rect(f/2B)$$

- Problem
  - Tail last for long time
    - Need more strict timing



$$h(t)=\frac{\sin(\pi t/T_s)}{\pi t/T_s}$$

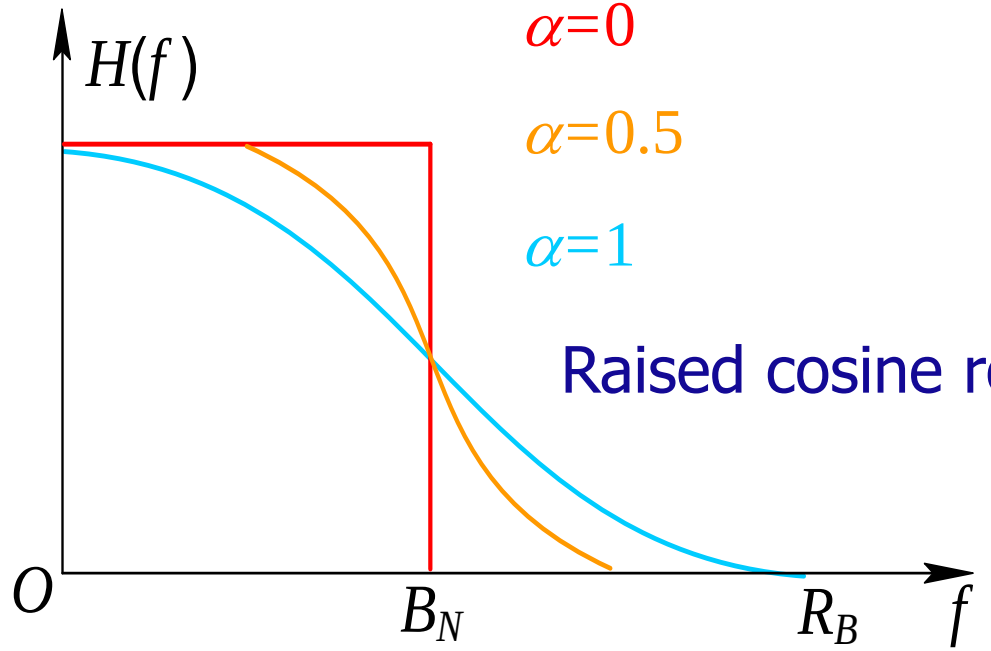
$$h(t)\propto t^{-1}$$

Complementary Symmetry Rolloff Lowpass System

- 特性：关于$f = B_N$ 奇对称
- 滚降系数

$$\alpha = \frac{B - B_N}{B_N}$$

  - $0 \le \alpha \le 1$
- 传输带宽 $B = (1+\alpha)B_N$
- 频带利用率 $\eta = 2/(1+\alpha)$



Raised cosine rolloff filter

- 常用：升余弦滚降系统
  - 传递函数为半个周期的余弦信号叠加直流分量

Raised Cosine Rolloff Filter

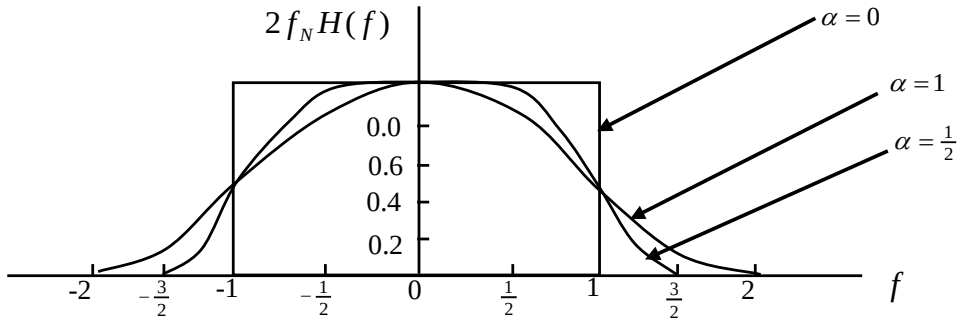
$$H(\omega)=\begin{cases} T_s, & 0\le|\omega|<\dfrac{(1-\alpha)\pi}{T_s} \\ \dfrac{T_s}{2}\left[1+\sin\dfrac{T_s}{2\alpha}\left(\dfrac{\pi}{T_s}-\omega\right)\right], & \dfrac{(1-\alpha)\pi}{T_s}<|\omega|\le\dfrac{(1+\alpha)\pi}{T_s} \\ 0, & |\omega|>\dfrac{(1+\alpha)\pi}{T_s} \end{cases}$$

$$h(t)=\frac{\sin\dfrac{\pi t}{T_s}}{\dfrac{\pi t}{T_s}}\cdot\frac{\cos\dfrac{\pi t}{T_s}}{1-\dfrac{4\alpha^2t^2}{T_s^2}}$$

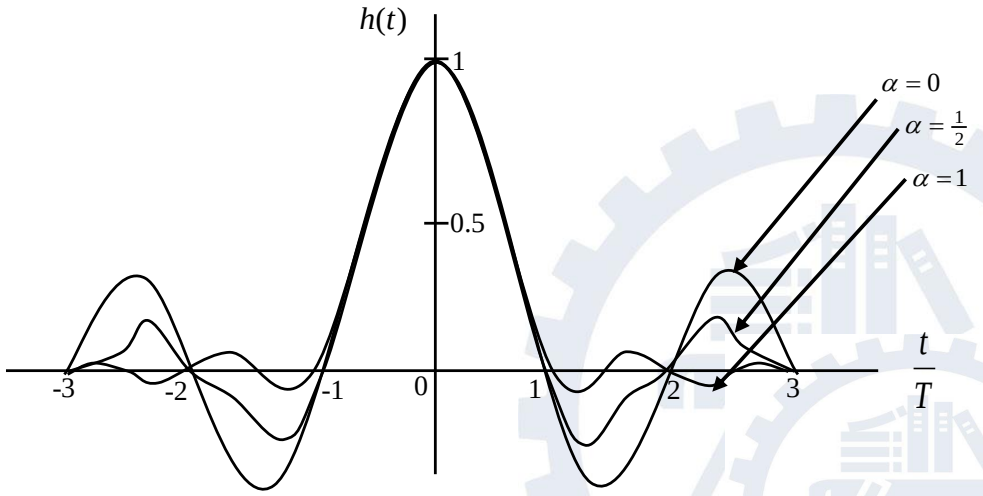
$h(t)\propto t^{-3}$

fast attenuation

α =1 full rolloff



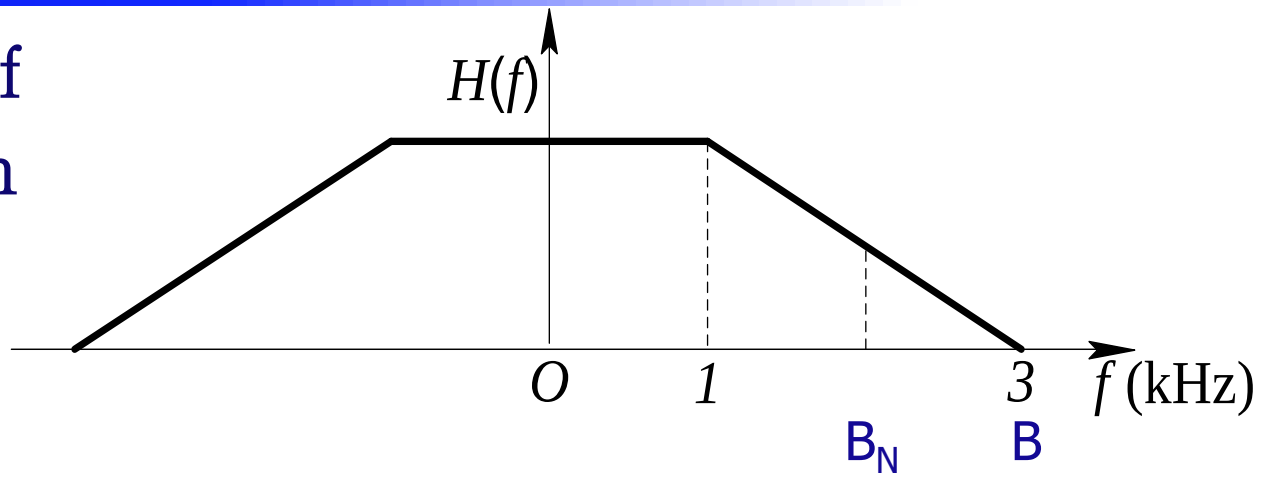
(a) 传递函数



(b) 单位冲激响应

Example

Transmission characteristics of binary digital baseband system are depicted as right figure.



Please calculate:

(1) B_N , R_B , B , α , η

(2) For 4-ary system, please recalculate

The rolloff part of $H(f)$ is complementary symmetry at B_N

$$B_N = \frac{R_B}{2} \rightarrow R_B = 2B_N$$

$$B = (1+\alpha) B_N \rightarrow \alpha = \frac{B - B_N}{B_N}$$

$$\eta = \frac{R_b}{B}$$

$$R_b = R_B \log_2 M$$

$$M = 2 \qquad M = 4$$

M-ary Digital Baseband Transmission System

- symbol period $T_{s(ymbol)}$ = sampling decision period $T_{s(ampling)}$
- symbol rate R_B = sampling decision rate f_s
- Nyquist bandwidth $B_N = R_B/2$
- transmission bandwidth $B = (1+\alpha)B_N$
- bandwidth efficiency $B = R_b/B = 2\cdot\log_2 M/(1+\alpha)$ (bit/s·Hz)

M levels denote *M* symbols

Summary of Baseband Transmission System under 1st Nyquist Criterion

- 奈氏第一准则：若系统传递函数 $H(f)$ 满足等效低通特性，以$R_B = 2B_N$的码率发送输入信号，则系统输出信号在抽样时刻kT_s上不存在ISI。

implementation	attenuation speed of $h(t)$ tail	effective bandwidth (Hz)	bandwidth efficiency (bit/s·Hz)
Ideal lowpass	t^{-1} slow	B_N	2
Rolloff lowpass	t^{-3} fast	$B_N(1+\alpha)$	$2/(1+\alpha)$

I want both fish and bear' paw, what can I do?



5.3.4 SECOND NYQUIST CRITERION



Waveform of impulse response

- Observation
  - Two **sin *x* / *x*** waveform of one symbol away have opposite polar in tail
  - Summation of such two waves can get a waveform with fast attenuated tail

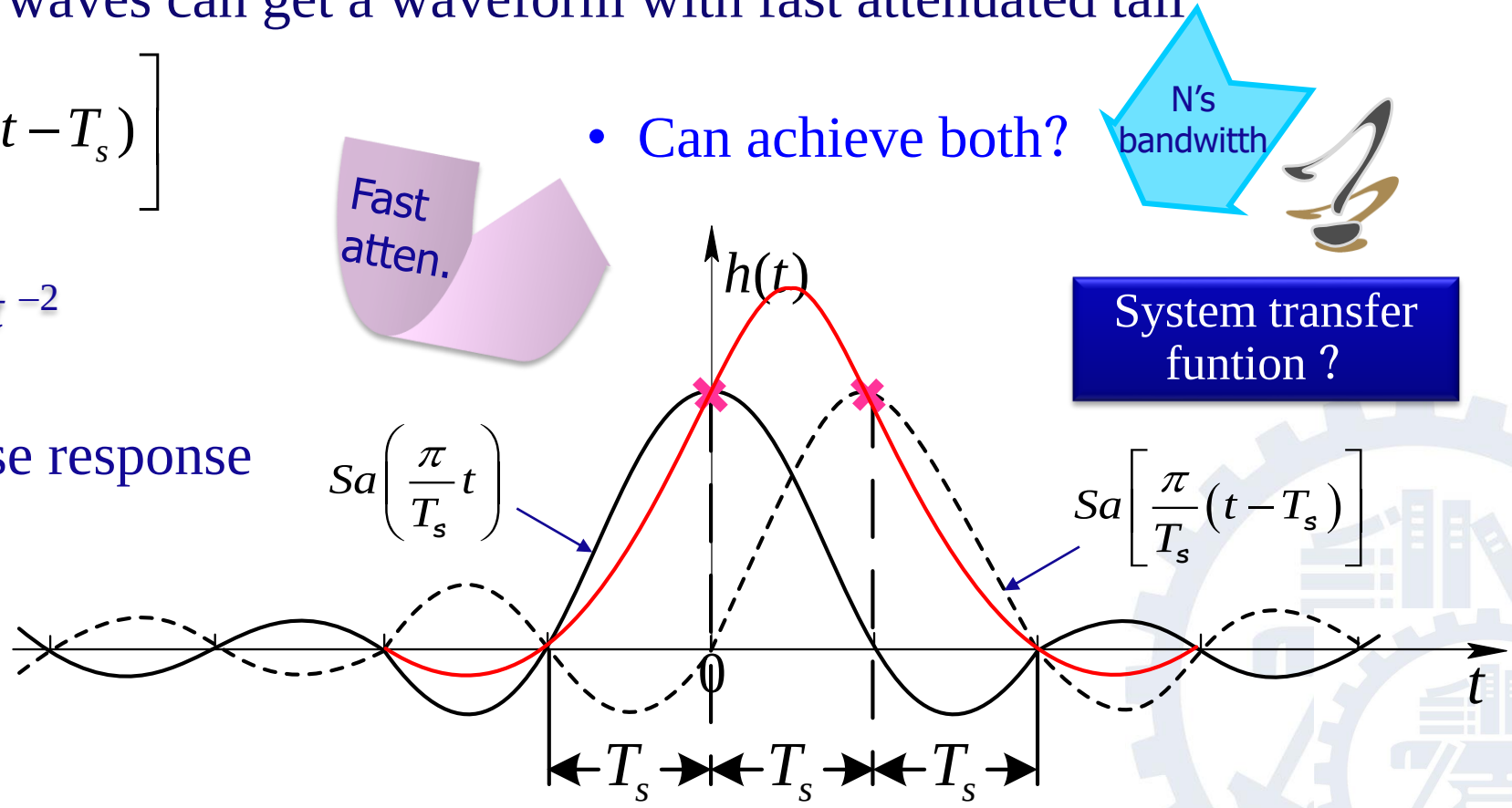
$$h(t) = Sa\left[\frac{\pi}{T_s}t\right] + Sa\left[\frac{\pi}{T_s}(t-T_s)\right]$$

$$= \frac{T_s^2 \sin(\pi t/T_s)}{\pi t(T_s - t)} \quad \propto t^{-2}$$

- Take it as system impulse response

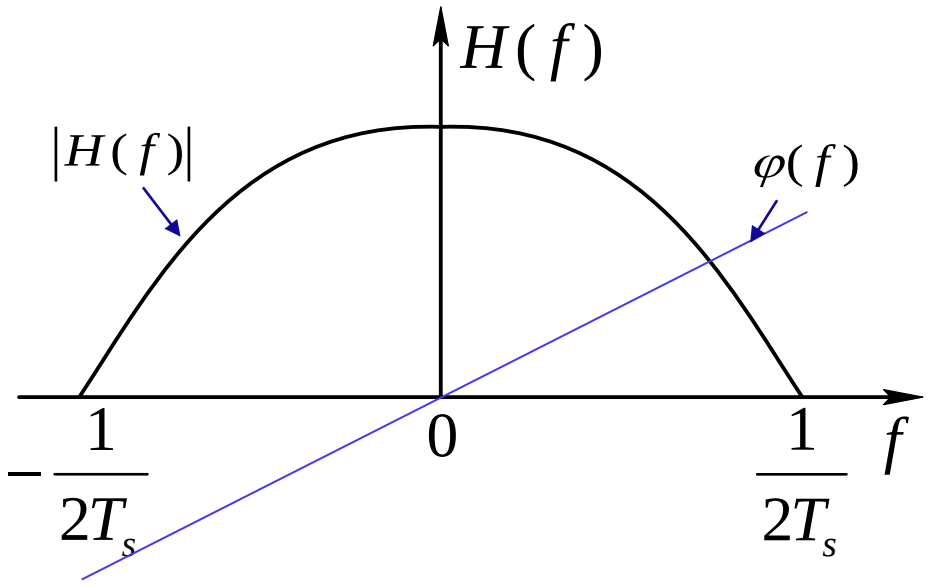
$$h(kT_s) = \begin{cases} C, & k = 0,1 \\ 0, & k \neq 0,1 \end{cases}$$

- Can achieve both?



System Transfer Function

$$H(f)=\begin{cases}2T_s\cos(\pi fT_s)\mathrm{e}^{-\pi fT_s} & |f|\le\dfrac{1}{2T_s}\\ 0 & |f|>\dfrac{1}{2T_s}\end{cases}$$



Unit Impulse Response

$$h(t)=Sa\left[\frac{\pi}{T_s}t\right]+Sa\left[\frac{\pi}{T_s}(t-T_s)\right]$$

$$=\frac{T_s^2\sin(\pi t/T_s)}{\pi t(T_s-t)}$$

N's bandwidth

$$B=B_N=1/2T_s=R_B/2$$

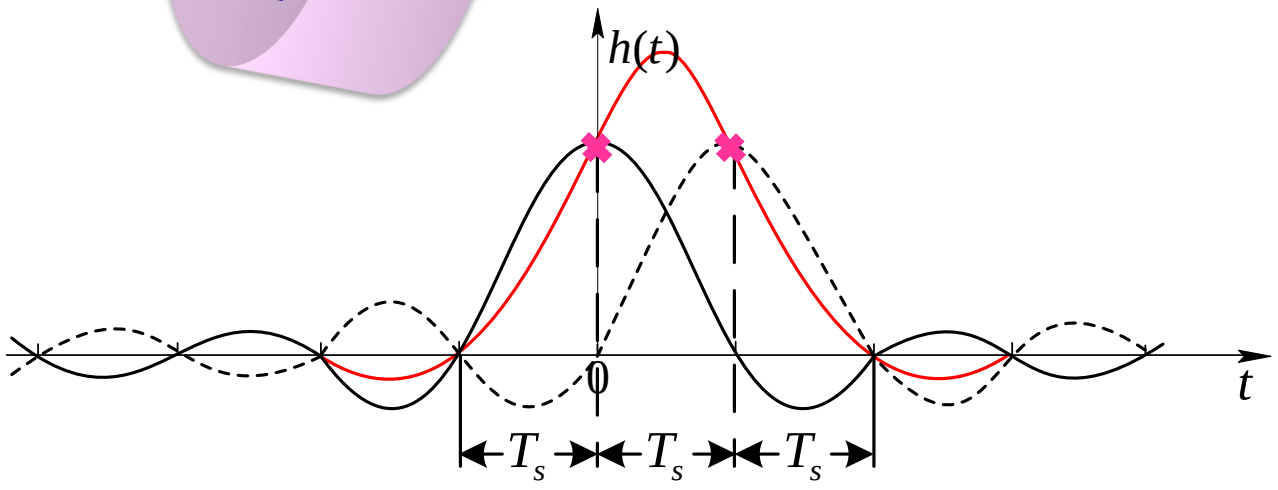
Get Both Fish and Bear's Paw

$$h(kT_s)=\begin{cases} C, & k=0,1 \\ 0, & k\neq 0,1 \end{cases}$$

2nd Nyquist Criterion

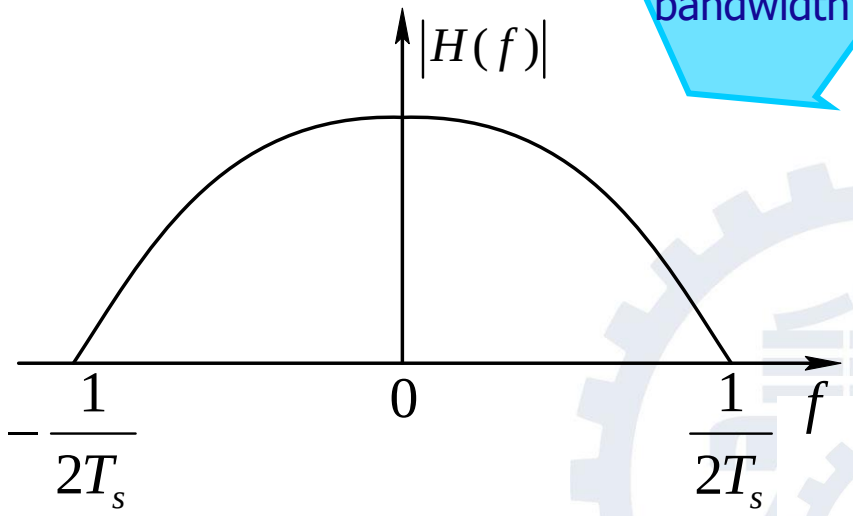
Fast atten.

$h(t)\propto t^{-2}$



$B = B_N = 1/2T_s = R_B/2$

N's bandwidth

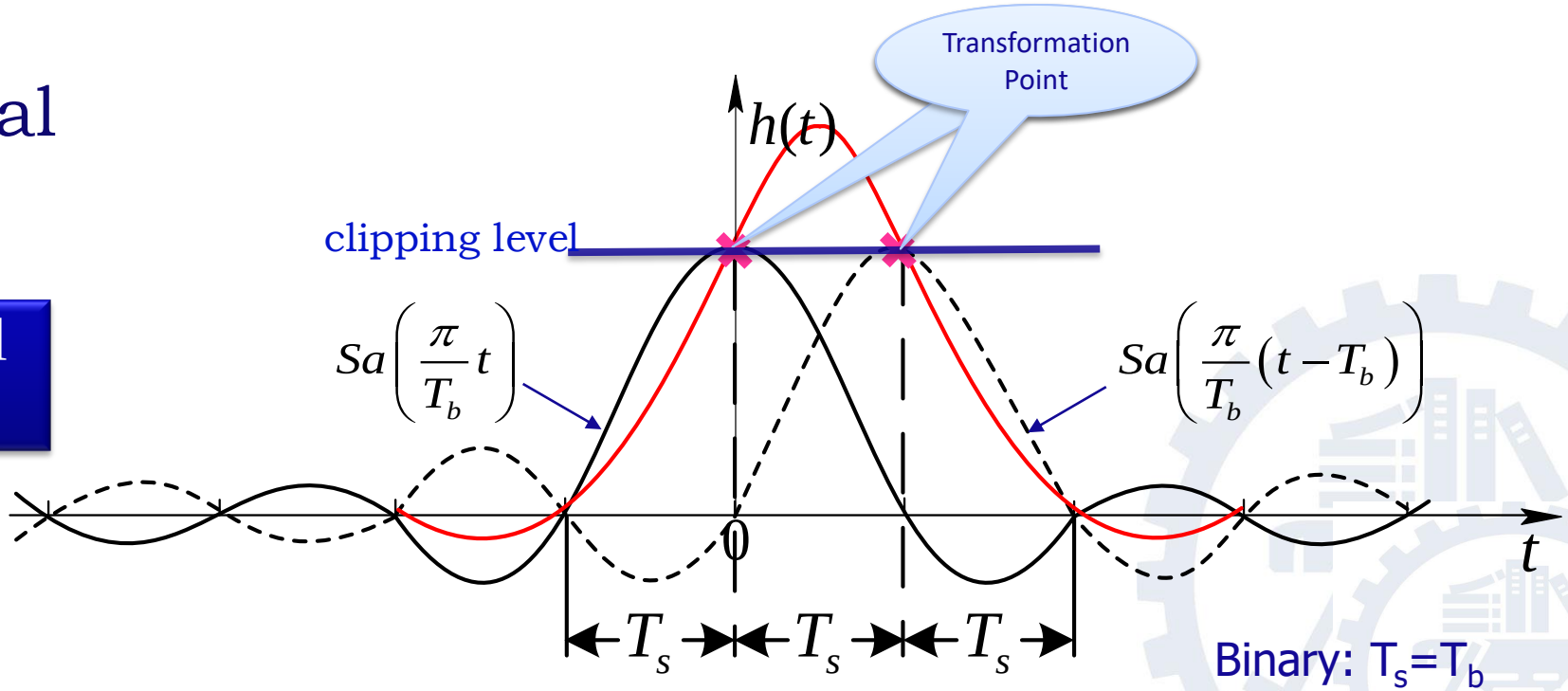


2nd Nyquist Criterion

- Interference only occurs between adjacent symbols
- At each sampling moment, system only responses partial symbols —— Partial Response

$$h(kT_s)=\begin{cases} C, & k=0,1 \\ 0, & k\neq 0,1 \end{cases}$$

How to design partial response system?



Binary: $T_s=T_b$

5.3.5 PARTIAL RESPONSE SYSTEM



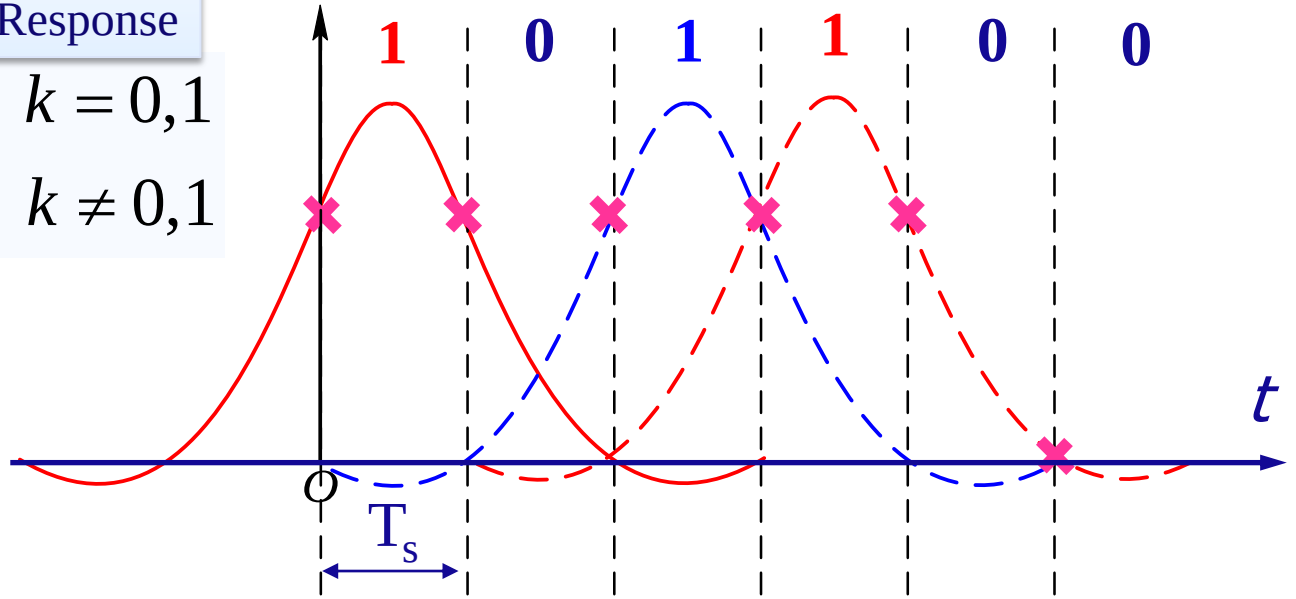
Correlation Coding

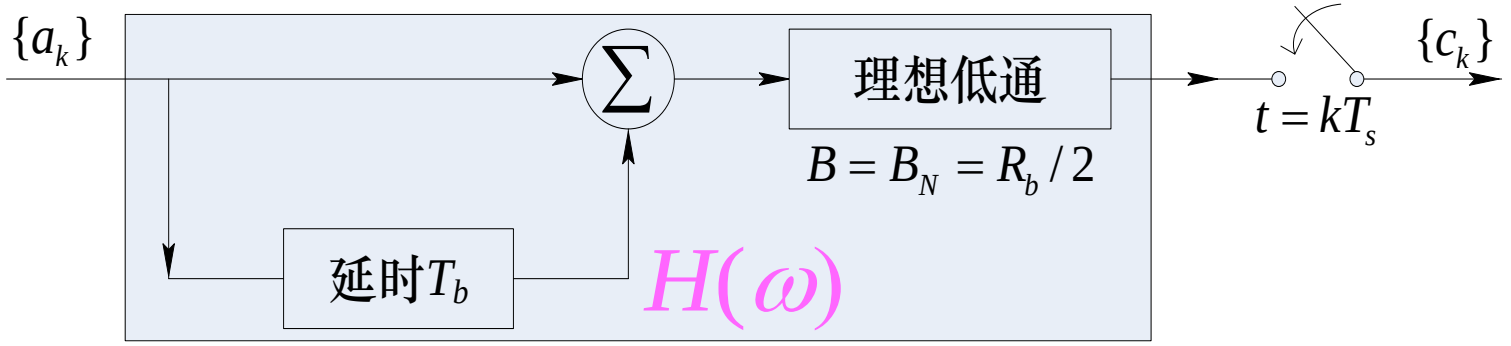
- Cor. C + Ideal LPF

$$c_k = a_k + a_{k-1} \Leftrightarrow h(kT_s)=\begin{cases} C, & k=0,1 \\ 0, & k\neq 0,1 \end{cases}$$

2nd Nyquist Criterion
System Impulse Response

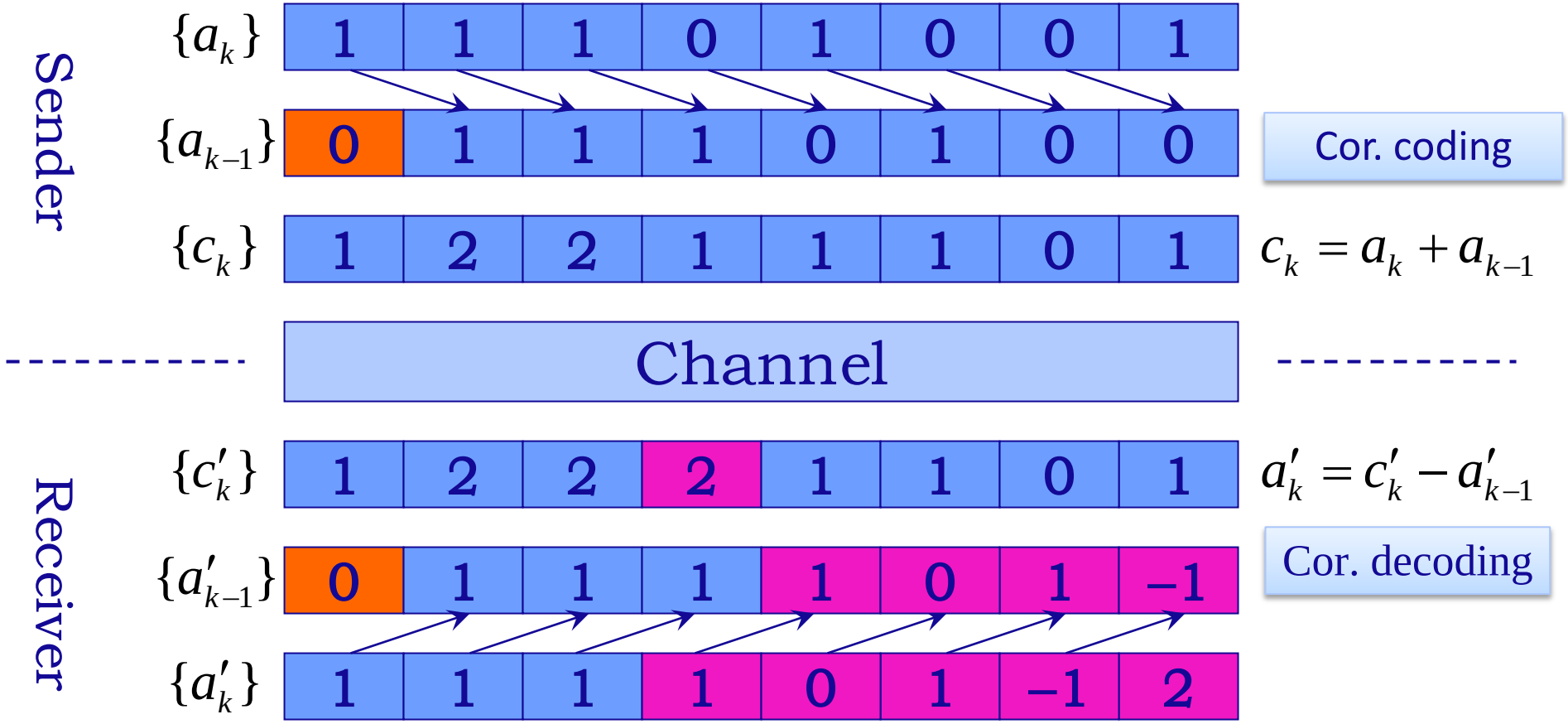
  - possible values of a_k: 0、1
  - possible values of c_k: 0、1、2 (three levels)





Error Propagation

- Error decision of one symbol could influence subsequent decisions

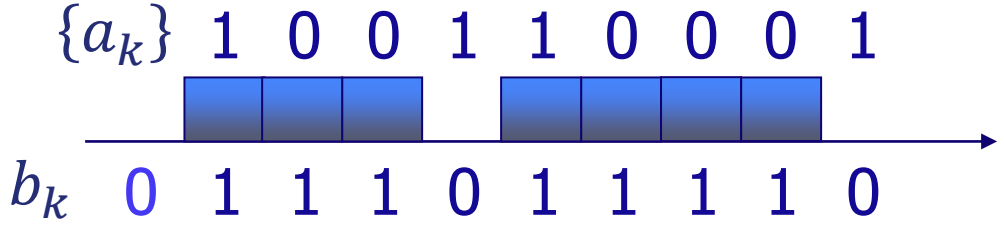


Solution

?

Differential Coding

- mark differential code



1 ~ change

$$b_k = a_k \oplus b_{k\text{-}1}$$

$$a_k = b_k \oplus b_{k\text{-}1}$$

Sender: Precoding + Cor. coding

Precoding: abs. code $a_k \rightarrow$ cor. code b_k

$$b_k = a_k \oplus b_{k-1}$$

Cor. coding: cor. Code $b_k \rightarrow$ partial resp. C_k

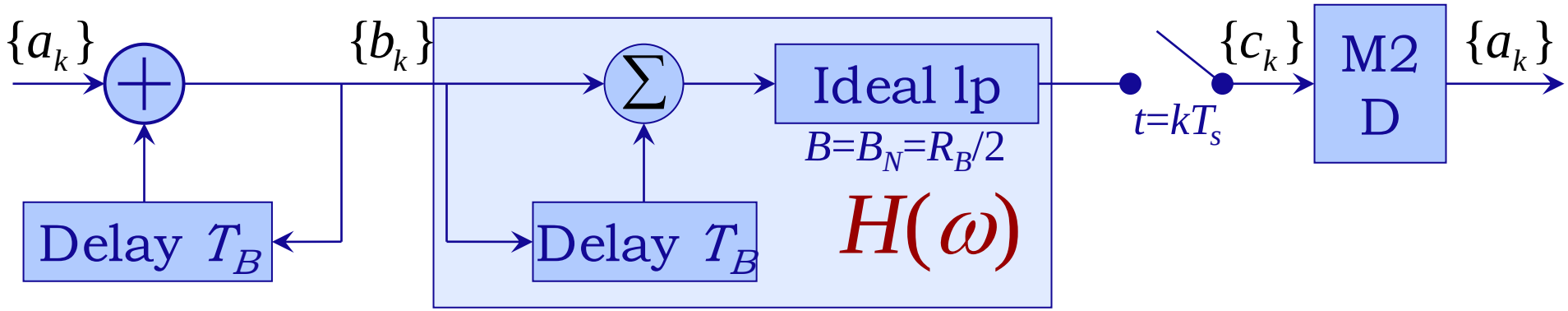
$$C_k = b_k + b_{k-1}$$

Receiver: modulo-2 decision

$$[C_k]_{\text{mod2}} = [b_k + b_{k-1}]_{\text{mod2}}$$
$$= b_k \oplus b_{k-1}$$
$$= a_k$$

Ignore channel noise

Class I Partial Response System



Sender: Precoding + Cor. coding

Precoding: abs. code $a_k \rightarrow$ cor. code b_k

$$b_k = a_k \oplus b_{k-1}$$

Cor. coding: cor. Code $b_k \rightarrow$ partial resp. C_k

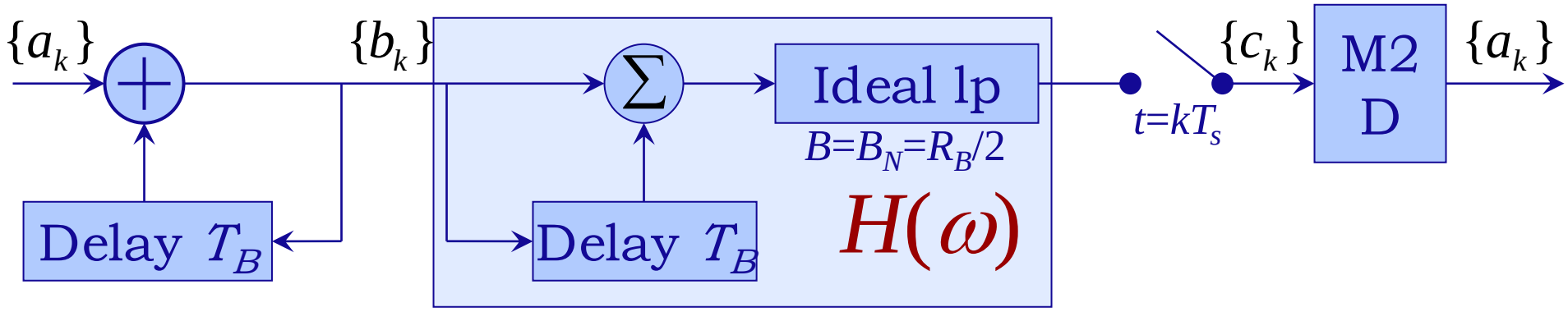
$$C_k = b_k + b_{k-1}$$

Receiver: modulo-2 decision

$$[C_k]_{\text{mod2}} = [b_k + b_{k-1}]_{\text{mod2}}$$
$$= b_k \oplus b_{k-1}$$
$$= a_k$$

Ignore channel noise

Class I Partial Response System



sender

precoding $b_k = a_k \oplus b_{k-1}$

cor. coding $C_k = b_k + b_{k-1}$

ignore noise $C'_k = C_k$

mod 2 decision $a_k = [C_k]_{\text{mod2}}$

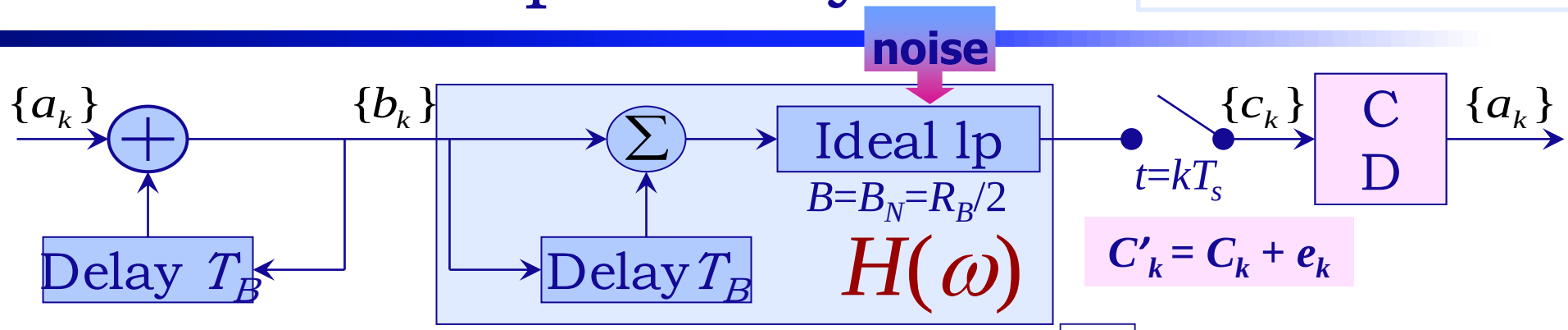
receiver

a_k		1	1	1	0	1	0	0	1
b_{k-1}		0	1	0	1	1	0	0	0
b_k	0	1	0	1	1	0	0	0	1
C_k		1	1	1	2	1	0	0	1
a_k		1	1	1	0	1	0	0	1

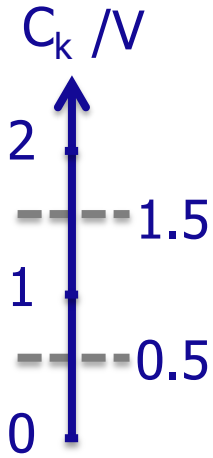
Each symbol is decided independently

Class I Partial Response System

Think of channel noise



$C'_k = C_k + e_k$



threshold level?

assume equal apriori probability

sender	a_k	1	1	1	0	1	0	0	1
	b_{k-1}	0	1	0	1	1	0	0	0
	b_k	1	0	1	1	0	0	0	1
	C_k	1	1	1	2	1	0	0	1
AWGN	e_k	-0.1	0.2	-0.3	-0.7	0.4	0.2	-0.2	-0.4
receiver	C'_k	0.9	1.2	0.7	1.3	1.4	0.2	-0.2	0.6
	a'_k	1	1	1	1	1	0	0	1

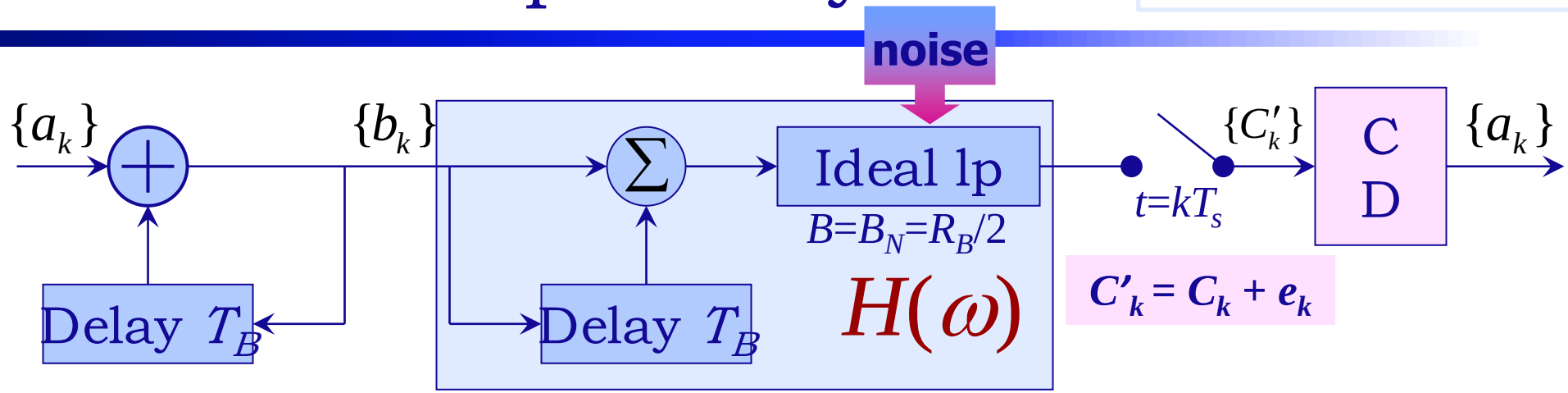
error

No error propagation

Receiver: compare decision
$C'_k \geq 1.5$ or $C'_k < 0.5$, "0";
$0.5 < C'_k < 1.5$, "1"

Class I Partial Response System

Think of channel noise



Sender: Precoding + Cor. coding

Precoding: abs. code $a_k \rightarrow$ cor. code b_k

$$b_k = a_k \oplus b_{k-1}$$

Cor. coding: cor. Code $b_k \rightarrow$ partial resp. C_k

$$C_k = b_k + b_{k-1}$$

Receiver: compare decision

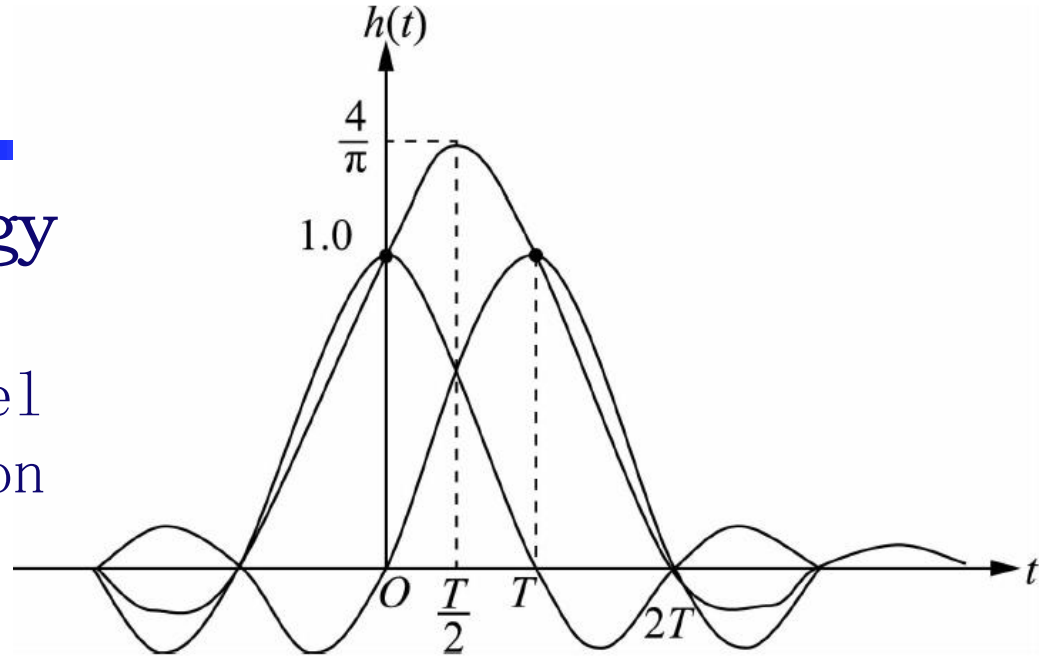
$C'_k \geq 1.5$ or $C'_k < 0.5$,
decide "0";

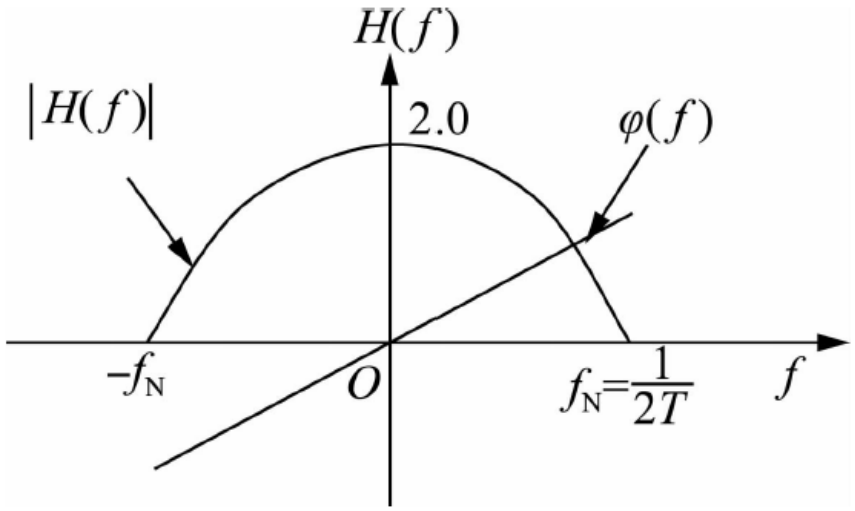
$0.5 < C'_k < 1.5$,
decide "1"

Drawbacks

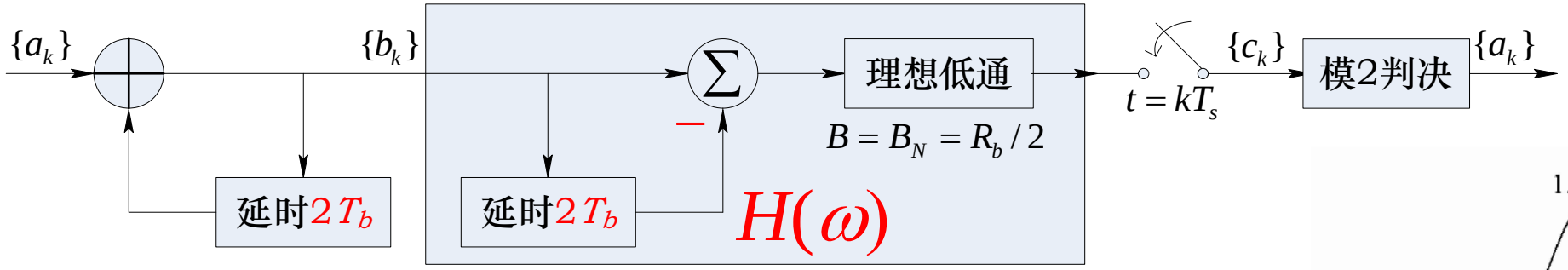
- Has DC component, and most energy concentrate at low frequency part
  - not suit low-frequency-restricted channel
  - not convenient for further SSB modulation
- Solution
  - “unipolar→polar” transfer
    - b_k : ±1, C_k level?
    - How to design decision rule when thinking of noise?
- Another solution
  - Modified duobinary signal

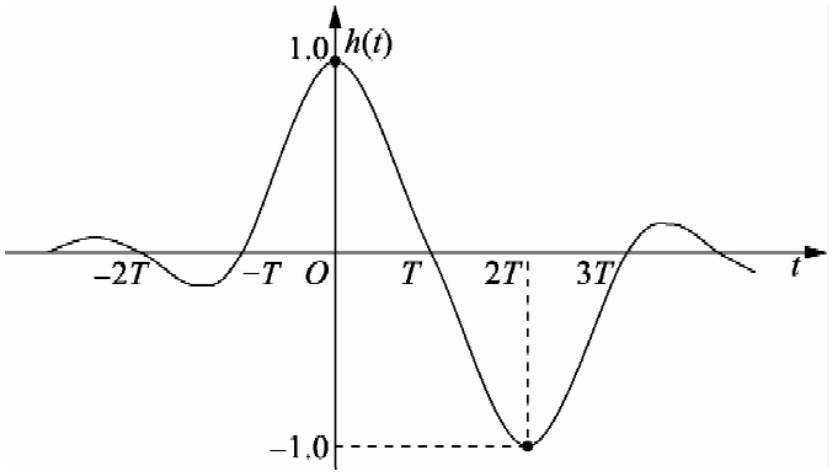
Class IV Partial Response System



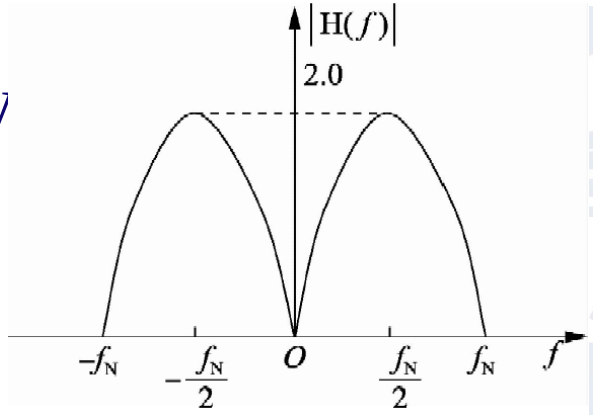


Class IV Partial Response System





- Delay unit changes from T_b to $2T_b$
  - Precoding: $b_k = a_k \oplus b_{k-2}$
  - Cor. coding: $c_k = b_k - b_{k-2}$
  - c_k: ± 1 and 0
- $H(\omega)$ has no DC component and few low frequency



Consider:
How to design decision rule when thinking of noise ?

Example

Suppose $\{a_k\}$=(10011010110), please write out class IV partial response code $\{c_k\}$



General Form of Partial Response System

- Response waveform is summation of N T_s-spaced $Sa(t)$ waves

$$h(t)=\sum_{n=0}^{N-1} W_n \mathrm{Sa}\left[\pi\left(\frac{t}{T}-n\right)\right] \quad \Leftrightarrow \quad H(\omega)=\begin{cases} T_s\sum_{k=1}^{N} W_k e^{-j\omega(k-1)T_s}, & |\omega|\le \dfrac{\pi}{T_s} \\ 0, & |\omega|>\dfrac{\pi}{T_s} \end{cases}$$

  - C_k level depends on a_k system and W_k value

$$C_k = W_0 b_k + W_1 b_{k-1} + \cdots + W_N b_{k-N}$$

  - Use precoding to avoid error propagation

$$a_k = W_0 b_k \oplus W_1 b_{k-1} \oplus \cdots \oplus W_N b_{k-N}$$

Class zero：Ideal lowpas sys.

Partial Response System (Class I-V)

- Bandwidth: B_N
- Band. Effi.: 2bit/s/Hz
- Spectrum and symbol interference
  - Class I: has DC component, and most energy concentrate at low frequency part
  - Class I: has DC component and few low frequency

- C_k levels > a_k levels
  - A large SNR is required to yield the same average probability of symbol error in presence of noise.

类 别	N	W_0	W_1	W_2	W_3	W_4	相关编码规则
第 I 类	2	1	1				$C_k = b_k + b_{k-1}$
第 II 类	3	1	2	1			$C_k = b_k + 2b_{k-1} + b_{k-2}$
第 III 类	3	2	1	−1			$C_k = 2b_k + b_{k-1} - b_{k-2}$
第 IV 类	3	1	0	−1			$C_k = b_k - b_{k-2}$
第 V 类	5	−1	0	2	0	−1	$C_k = -b_k + 2b_{k-2} - b_{k-4}$

Consider: C_k level?

5.3.6 OPTIMIZATION OF BASEBAND TRANSMISSION



Optimization of Baseband Transmission

- As $H(f)$ already known, how to determine the transmit and receive filter?

$$H(f) = H_T(f) \cdot H_C(f) \cdot H_R(f) = |H(f)| e^{-j2\pi f t_0}$$

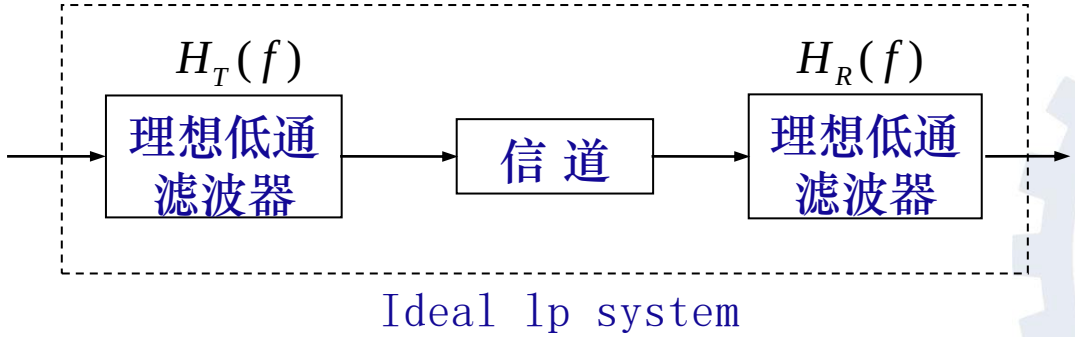
  - amp.-freq. $|H(f)| = |H_T(f)| \cdot |H_C(f)| \cdot |H_R(f)|$
  - phase-freq. $\phi_T(f) + \phi_C(f) + \phi_R(f) = -j2\pi f t_0$
  - Total delay $t_0 = t_T + t_C + t_R$
  - channel has been well equalized $H(f) = H_T(f) H_R(f) = |H(f)| e^{-j2\pi f (t_T + t_R)}$

- The overall transmission characteristic should be split evenly between the transmit and receive filter

$$H_T(f) = \sqrt{|H(f)|} e^{-j2\pi f t_T}$$

$$H_R(f) = \sqrt{|H(f)|} e^{-j2\pi f t_R}$$

Why?
You can find answer in Chap. 7 optimal reception



Ideal lp system

- Problem Analysis of Symbol Error Rate
- Optimum Threshold Level
- Symbol Error Rate with Optimum Threshold Level

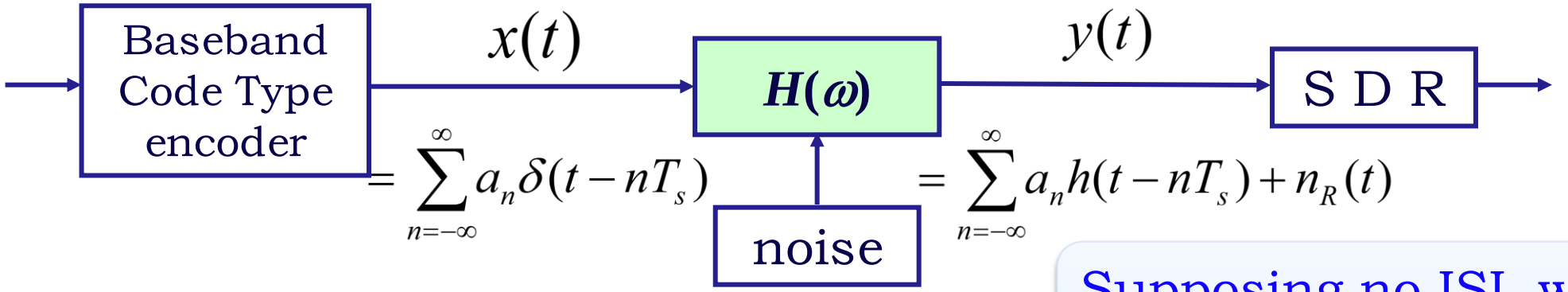
5.4 ERROR PERFORMANCE

5.4.1 PROBLEM ANALYSIS OF SYMBOL ERROR RATE



Problem Description

- Analyzing model of baseband transmission characteristic with noise



Supposing no ISI, what is the noise influence?

kT_s moment $\quad y(kT_s) = \sum_{n=-\infty}^{+\infty} a_n h[(k-n)T_s] + n_R(kT_s)$

$$= a_k h(0) + \sum_{n \neq k} a_n h[(k-n)T_s] + n_R(kT_s)$$

sample value of kth symbol — ISI — random interference

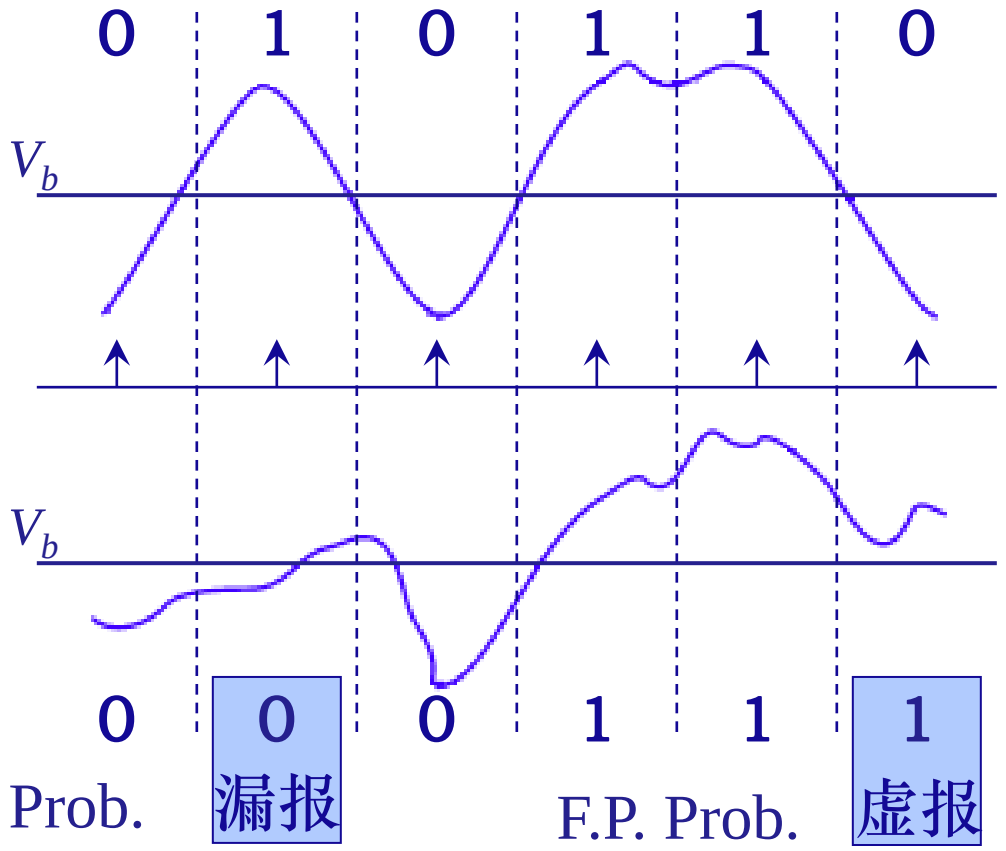
Wrong Decision: False Positive & False Negative

- Two kinds of mistakes: False Positive & False Negative

Take 2-level system for example

(a) Without noise

Input waveform of sample circuit



$y(kT_s)=a_k h(0)$

discrete continuous

$y(kT_s)=a_k h(0)+n_R(kT_s)$

Symbol-valued random variables

Receiving noise random varibiles

Introduce conditional probability to simplify the analysis

F.N. Prob. $P_{\text{Me}} = P(1)P(0|1)$

F.P. Prob. $P_{\text{Se}} = P(0)P(1|0)$

Two Kinds of Conditional Probability

- Sample value for single symbol (random variable)Y consists of two parts

$$Y = V + G$$

  - Disc. rand. Vari. with 2 possible values

$$V = \begin{cases} A_0 & P(0) \\ A_1 & P(1) \end{cases}$$

  - Continuous gaussian distribution

$$G \sim N(0, \sigma_n^2)$$

- Y ~ gaussian distribution $Y \sim N(V, \sigma_n^2)$
- Y have 2 kinds of conditional probability

$$p_0(y) = p(y \mid 0) = N(A_0, \sigma_n^2)$$

$$p_1(y) = p(y \mid 1) = N(A_1, \sigma_n^2)$$

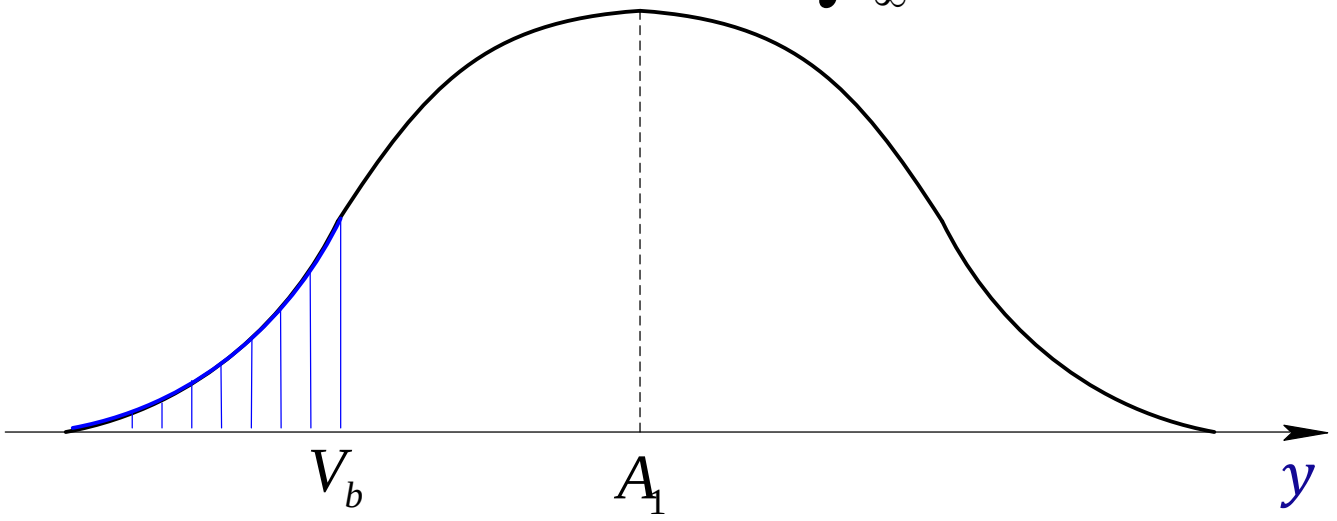
Probability of False Negative

- Decision threshold level V_b locates between two symbol level $A_0 < V_b < A_1$

- F.N. Prob.

$$P_{\text{Me}} = P(1)P(0|1) = P(1)P(y < V_b \mid 1)$$

$$= P(1)\int_{-\infty}^{V_b} N(A_1, \sigma_n^2)dy$$



$$p_1(y) = p(y \mid 1) = N(A_1, \sigma_n^2)$$

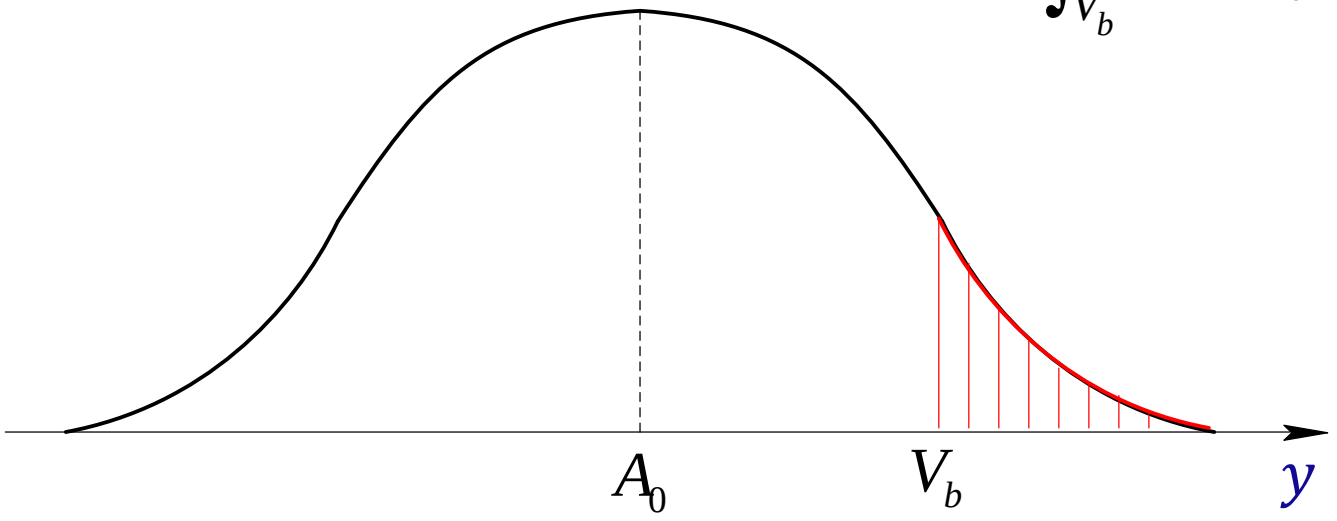
Probability of False Positive

- Decision threshold level V_b locates between two symbol level $A_0 < V_b < A_1$

- F. P. Prob.

$$P_{\text{Se}} = P(0)P(1|0) = P(0)P(y > V_b \mid 0)$$

$$= P(0)\int_{V_b}^{\infty} N(A_0, \sigma_n^2)dy$$



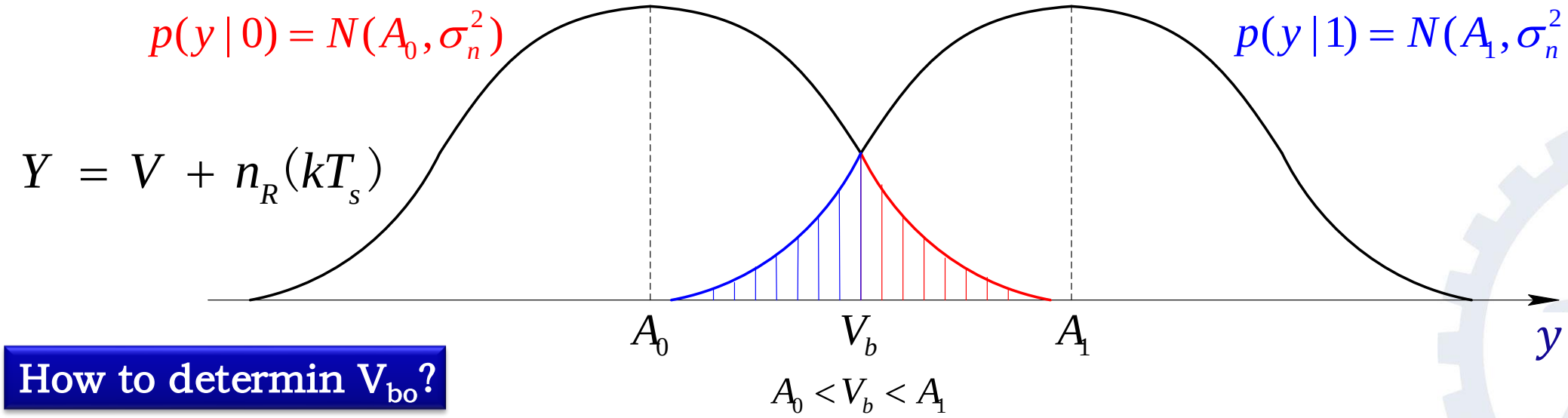
$$p_0(y) = p(y \mid 0) = N(A_0, \sigma_n^2)$$

Symbol Error Rate P_e

- P_e is a function of decision threshold level V_b
- For given SNR (for given A_0、A_1、σ_n)，V_b takes a level that minimize P_e

——the level is called optimum threshold level V_{bo}

$$P_e = P_{\text{Me}} + P_{\text{Se}} = P(1)\int_{-\infty}^{V_b} N(A_1, \sigma_n^2)dy + P(0)\int_{V_b}^{\infty} N(A_0, \sigma_n^2)dy$$



How to determin V_{bo}?

5.5.2 OPTIMUM THRESHOLD LEVEL



Equation of Optimum Threshold Level

$$P_e = P_{Me} + P_{Se} = P(1)\int_{-\infty}^{V_b} N(A_1,\sigma_n^2)dy + P(0)\int_{V_b}^{\infty} N(A_0,\sigma_n^2)dy$$

$$= P(1)\frac{1}{\sqrt{2\pi}\sigma_n}\int_{-\infty}^{V_b} e^{-\frac{(y-A_1)^2}{2\sigma_n^2}}dy + P(0)\frac{1}{\sqrt{2\pi}\sigma_n}\int_{V_b}^{\infty} e^{-\frac{(y-A_0)^2}{2\sigma_n^2}}dy$$

$$= P(1)\int_{-\infty}^{V_b} p_1(y)dy + P(0)\int_{V_b}^{\infty} p_0(y)dy$$

$$p_1(x) = p(x|1) = \frac{1}{\sqrt{2\pi}\sigma_n} e^{-\frac{(x-A_1)^2}{2\sigma_n^2}}$$

V_{bo} that minimize P_e should satisfies $\left.\frac{dP_e}{dV_b}\right|_{V_b=V_{bo}} = 0$

Using Leibniz Formula, we get $P(1)p_1(V_{bo}) = P(0)p_0(V_{bo})$

Equation of Optimum Threshold Level

Leibniz Formula

$$I(\alpha) = \int_{a(\alpha)}^{b(\alpha)} f(x,\alpha)dx$$

integration boundary is variable

$$\frac{dI(\alpha)}{d\alpha} = \int_{a(\alpha)}^{b(\alpha)} \frac{df(x,\alpha)}{d\alpha}dx$$

$$= f(b(\alpha),\alpha)\frac{db(\alpha)}{d\alpha} - f(a(\alpha),\alpha)\frac{da(\alpha)}{d\alpha}$$

Optimum Threshold Level V_{bo}

Solve the equation $P(1)p_1(V_{bo}) = P(0)p_0(V_{bo})$

$$p_1(y) = p(y|1) = \frac{1}{\sqrt{2\pi}\sigma_n} e^{-\frac{(y-A_1)^2}{2\sigma_n^2}}$$

$$P(1)e^{-\frac{(V_{bo}-A_1)^2}{2\sigma_n^2}} = P(0)e^{-\frac{(V_{bo}-A_0)^2}{2\sigma_n^2}}$$

$$\ln P(1) - \frac{(V_{bo}-A_1)^2}{2\sigma_n^2} = \ln P(0) - \frac{(V_{bo}-A_0)^2}{2\sigma_n^2}$$

$$(V_{bo}-A_0)^2 - (V_{bo}-A_1)^2 = 2\sigma_n^2 \ln\frac{P(0)}{P(1)}$$

Optimum Threshold Level V_{bo}

$$V_{bo} = \frac{A_1+A_0}{2} + \frac{\sigma_n^2}{A_1-A_0}\ln\frac{P(0)}{P(1)}$$

$$V_{bo} = \frac{A_1+A_0}{2} \qquad P(1) = P(0)$$

- Unipolar(A_1=A, A_0=0)
- V_{bo}=$A/2$
- Polar(A_1=A, A_0= $-A$)
- V_{bo}=0

5.4.3 SYMBOL ERROR RATE WITH OPTIMUM THRESHOLD LEVEL



Symbol Error Rate with Optimum Threshold Level

$$P_e = P_{Me} + P_{Se} = P(1)\int_{-\infty}^{V_b} N(A_1, \sigma_n^2)dy + P(0)\int_{V_b}^{\infty} N(A_0, \sigma_n^2)dy$$

$$= \frac{1}{2}\frac{1}{\sqrt{2\pi}\sigma_n}\left[\int_{-\infty}^{V_{bo}} \mathrm{e}^{-\frac{(y-A_1)^2}{2\sigma_n^2}}\,dy + \int_{V_{bo}}^{\infty} \mathrm{e}^{-\frac{(y-A_0)^2}{2\sigma_n^2}}\,dy\right]$$

$$= \frac{1}{2\sqrt{\pi}}\left[\int_{-\infty}^{\frac{V_{bo}-A_1}{\sqrt{2\sigma_n^2}}} \mathrm{e}^{-y^2}\,dy + \int_{\frac{V_{bo}-A_0}{\sqrt{2\sigma_n^2}}}^{\infty} \mathrm{e}^{-y^2}\,dy\right]$$

$$= \frac{1}{2\sqrt{\pi}}\left[\int_{-\infty}^{-y_0} \mathrm{e}^{-y^2}\,dy + \int_{y_0}^{\infty} \mathrm{e}^{-y^2}\,dy\right] = \frac{1}{\sqrt{\pi}}\int_{y_0}^{\infty} \mathrm{e}^{-y^2}\,dy$$

$$= \frac{1}{2}\mathrm{erfc}(y_0) = \frac{1}{2}\mathrm{erfc}\left(\frac{A}{2\sqrt{2\sigma_n^2}}\right) = Q\left(\frac{A}{2\sigma_n}\right)$$

$$V_{bo} = \frac{A_1 + A_0}{2}$$

0、1 take equal prob.

$$y = \frac{y - A_i}{\sqrt{2\sigma_n^2}} \quad i = 0,1$$

peak-to-peak value

$$A = A_1 - A_0$$

$$\frac{V_{bo} - A_0}{\sqrt{2\sigma_n^2}} = \frac{A}{2\sqrt{2\sigma_n^2}} = y_0$$

$$\frac{V_{bo} - A_1}{\sqrt{2\sigma_n^2}} = \frac{-A}{2\sqrt{2\sigma_n^2}} = -y_0$$

e^{-y^2} symmetry

$$\mathrm{erfc}(x) = \frac{2}{\sqrt{\pi}}\int_x^{\infty} e^{-z^2}\,dz$$

$$Q(x) = \frac{1}{2}erfc\left(\frac{x}{\sqrt{2}}\right)$$

Symbol Error Rate with Optimum Threshold Level

$$P_e = \frac{1}{2} erfc\left(\frac{\sqrt{\gamma}}{2}\right)$$

$$\gamma = \frac{A^2}{2\sigma_n^2}$$

signal peak power

noise average power

"Input SNR"

- P_e only relates to γ, having nothing to do with signal form
- more lager γ, more lower P_e

Discussion: ISI influence?
Optimum threshold level with ISI?

$$A = A_1 - A_0$$

$$P_e = \frac{1}{2} erfc\left(\frac{A}{2\sqrt{2}\sigma_n}\right)$$

$$V_{bo} = \frac{A_1 + A_0}{2}$$

Average power of digital baseband signal

$$S = pS_1 + (1-p)S_0 \quad S_1 = \frac{1}{T}\int_{-T/2}^{T/2} s_1^2(t)dt$$

- Given peak-peak value A and same $\mathbf{P_e}$
  - Unipolar NRZ(A_1=A, A_0=0)
    - Average power $S=A^2/2$
  - Bipolar NRZ (A_1=A/2, A_0= – A/2)
    - Average power $S=A^2/4$
- **Given average power, Bipolar has lower $\mathbf{P_e}$**

Example

A computer produces bipolar pulse sequence with bit rate R_b=1Mbps, transmitted through a channel where noise power spectrum n_0=4×10^{-8} W/Hz. Please calculate the power and amplitude of the signal to ensure that symbol error rate is not more than 3×10^{-5}.

$$P_e = \frac{1}{2} erfc\left(\frac{\sqrt{\gamma}}{2}\right)$$

$$\gamma = \frac{A^2}{2\sigma_n^2} \qquad A = A_1 - A_0 \qquad N = \sigma_n^2 = n_0 B \qquad B = B_N = \frac{R_B}{2}$$

- Eye Pattern
- Interpretation of Eye Pattern

5.5 EYE PATTERN

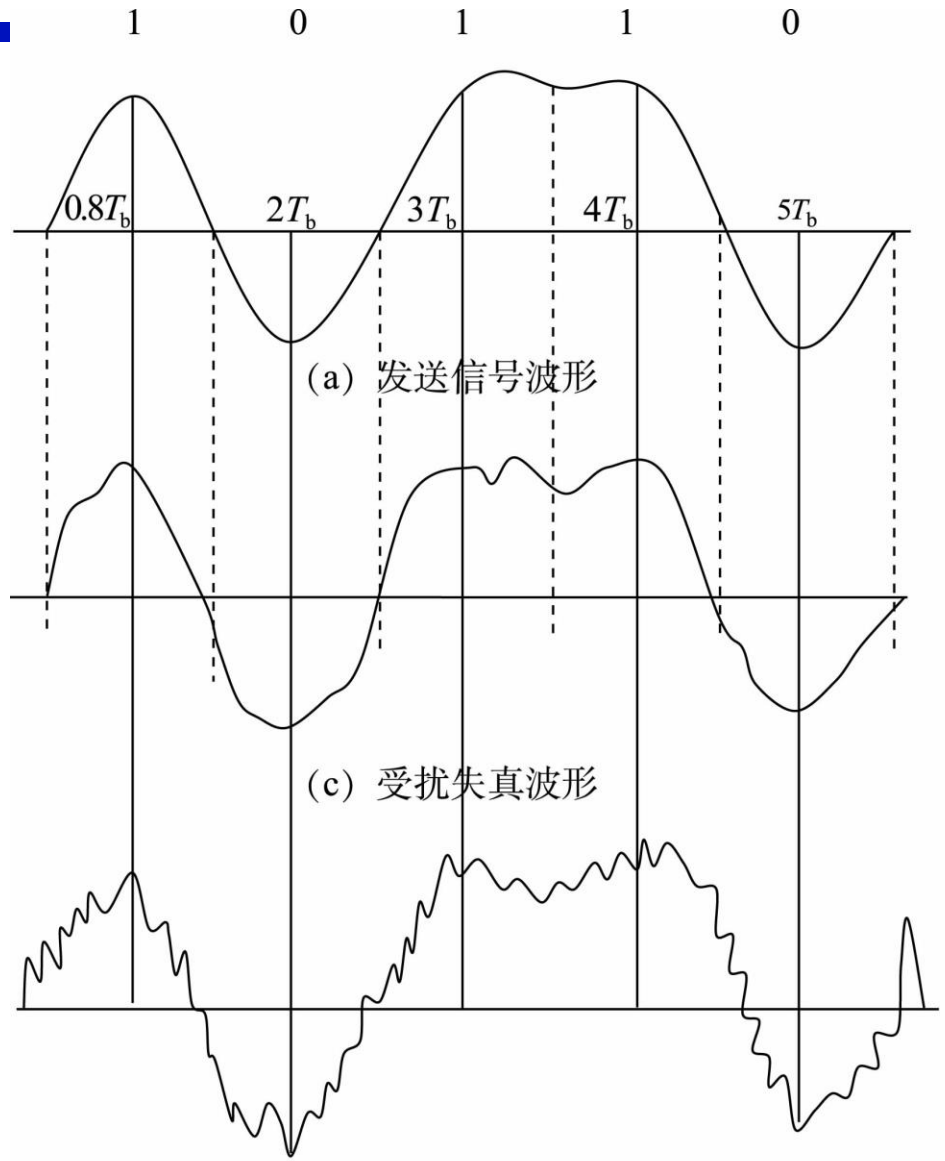
Why Check Eye Pattern?

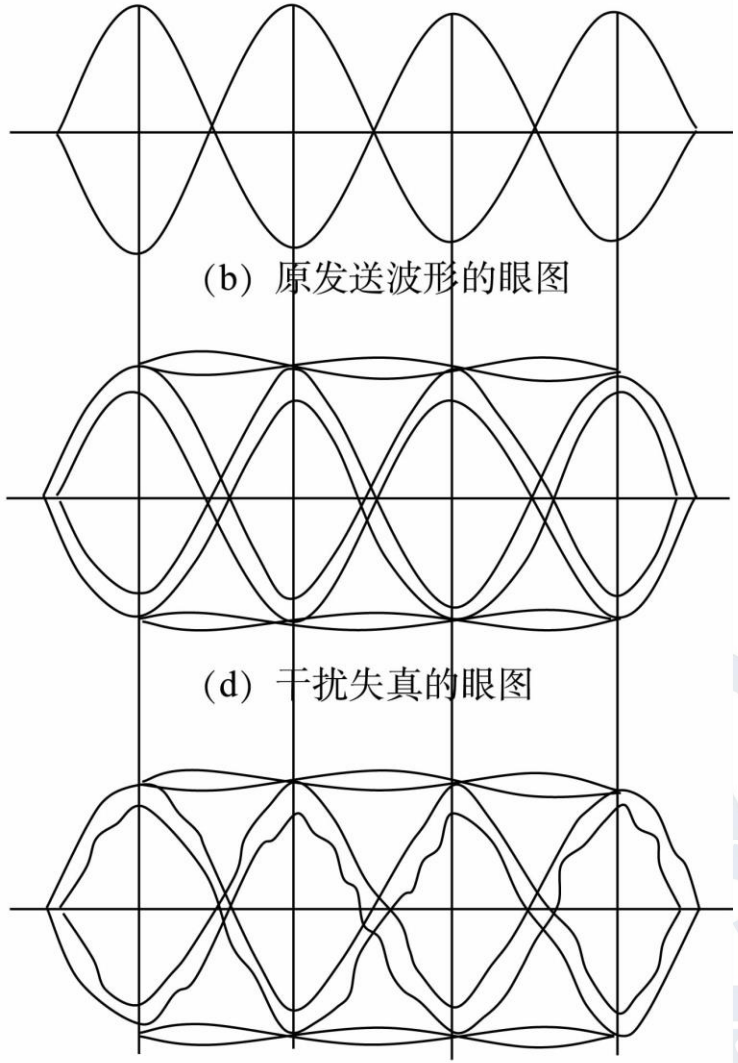
- Eye pattern is a means of evaluating the quality of a received “digital waveform”
  - Quality is meant the ability to correctly recover symbols and timing
  - The received signal could be examined at the input to a digital receiver or at some stage within the receiver before the decision stage
- Eye pattern reveal the impact of ISI and noise
- Two major issues are 1) sample value variation, and 2) jitter and sensitivity of sampling instant
- Eye pattern reveals issues of both
- Eye pattern can also give an estimate of achievable BER

What is eye pattern?

- Set the sweep period of oscilloscope to integer times of symbol period T, the resulting oscilloscope display is something like human eye, so is called eye pattern

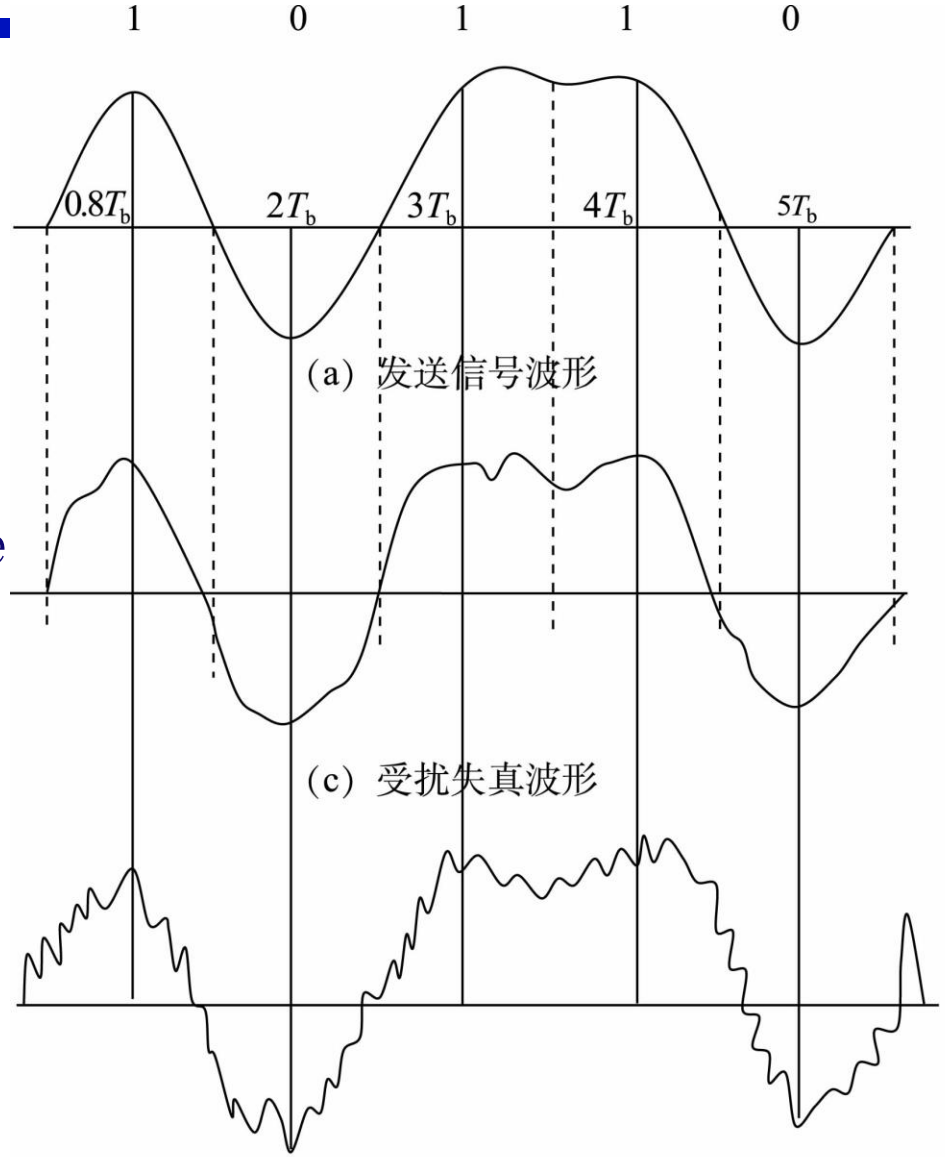
- The width of eye opening reveals
  - the influence of ISI and additive noise



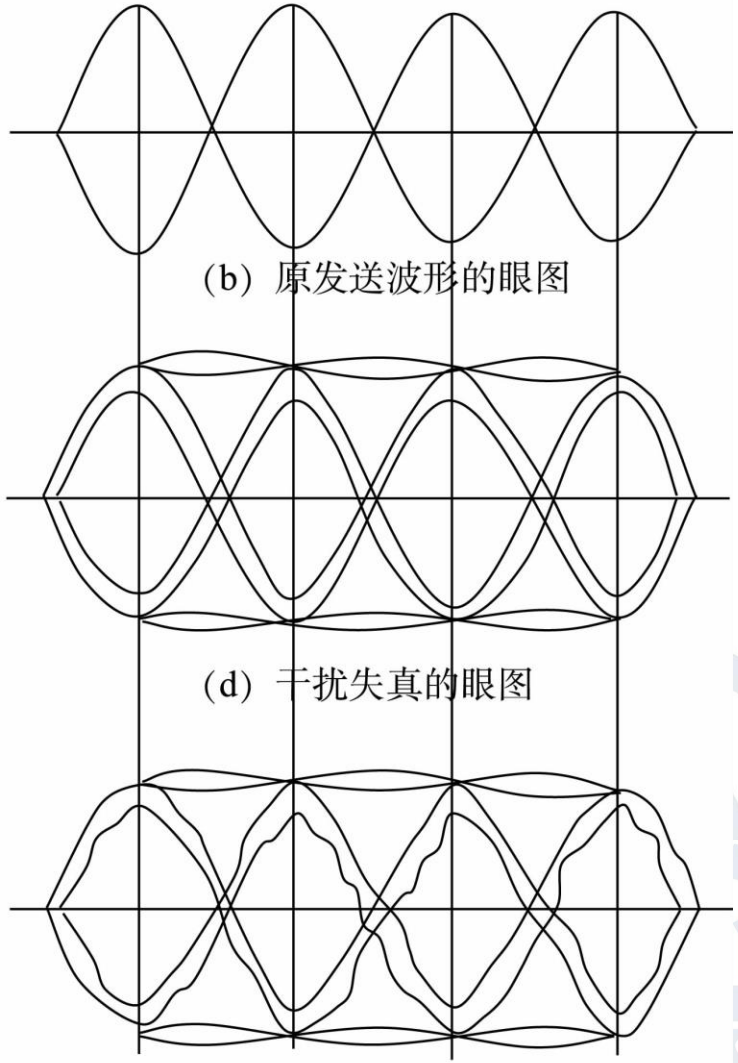


The Width of Eye Opening

- Clear opening
  - No ISI and noise
- Partial opening or closed
  - Have ISI and noise
  - Dispersive trace make the eye closed

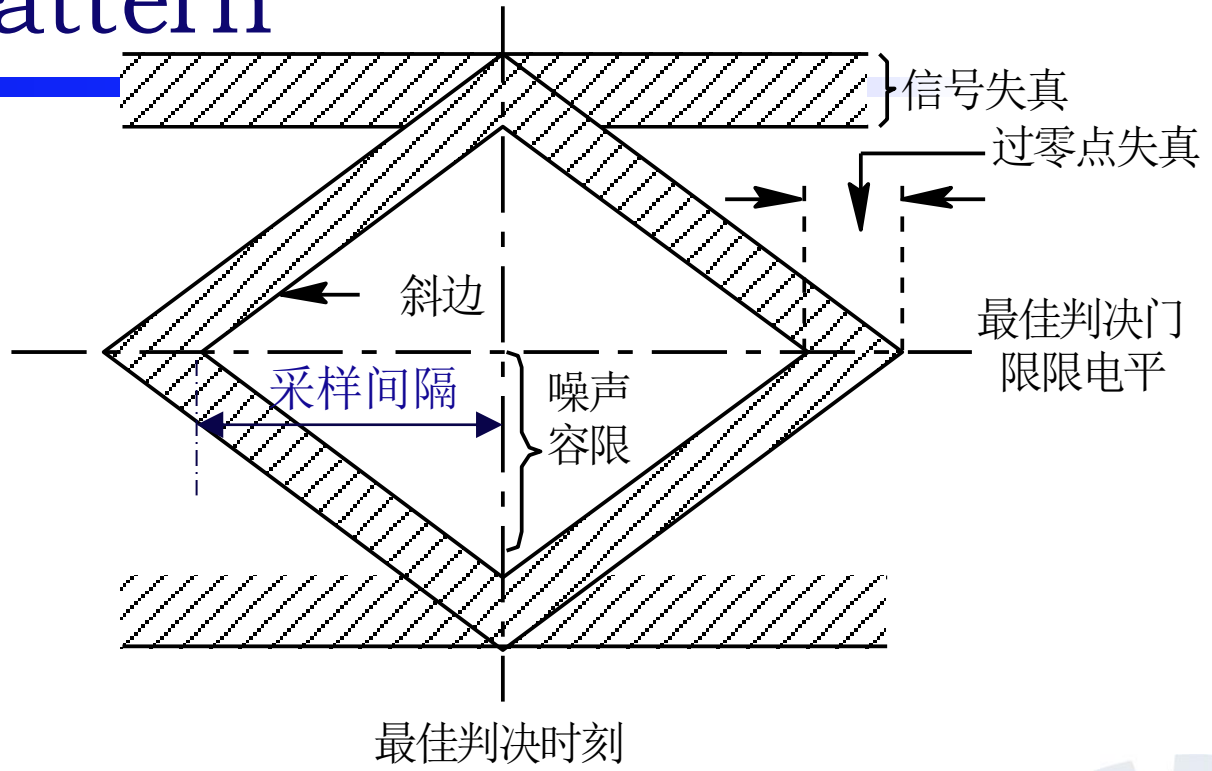


Setting the sweep period to 4T results 4 opening eyes



Interpretation of Eye Pattern

- Optimal sampling time
  - Maximum eye opening
- Noise margin
  - Height of eye opening
- Sample interval
  - Width of eye opening
- Slope
  - High slope means high sensitivity to timing error
- Distortion of zero crossing
  - Distortion of decision threshold, sensitive to a synchronization error
- Distortion of sampling
  - Amount of distortion at sampling instant, relates to signal SNR



Consider: what is the influence of ISI for sampling instant and decision threshold?

Observe: eye pattern of M-ary signal?

- Equalization: Concept
- Transversal Filter

5.6 TIME-DOMAIN EQUALIZATION

Equalization

- Well-designed transmit and receive filters can eliminate the influence of ISI and noise
- Problem: channels are nonideal
  - Filters may be not accurate enough, and channel characteristics are either unknown or change with time
- Solution: equalization
  - Add an adjustable compensative filter between receive filter and sample decision
  - Adjust the filter to compensate characteristics of the channel or the whole transmission system, thus decrease the influence of ISI

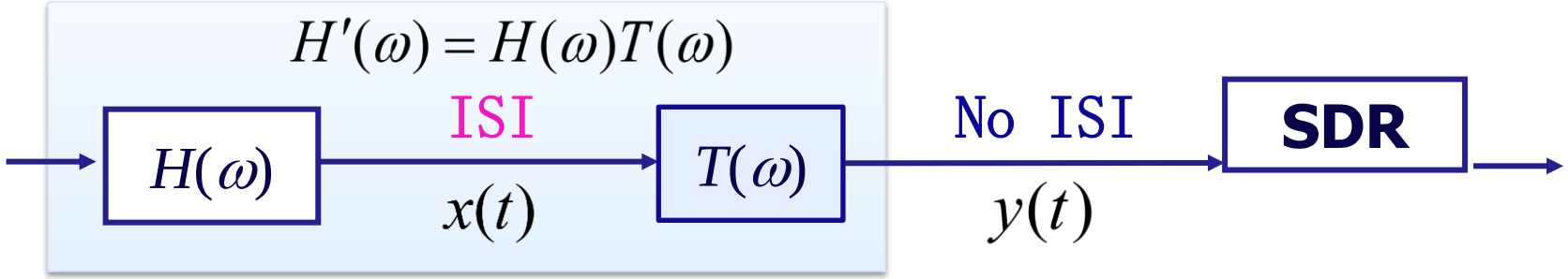
Category of Equalization

- Frequency-domain: frequency-response compensation
- Time-domain: receive waveform compensation

- Linear: equalizer is linear system
- Nonlinear: equalizer is nonlinear system

- Pre-equalization: learning/training
- Adaptive equalization: track time variations in the channel response

Time-domain Equalization



$H'(\omega)$ *satisfies* the Nyquist criterion

$$\sum_{n=-\infty}^{\infty} H'(\omega - \frac{2n\pi}{T_s}) = T_s \qquad |\omega| \le \frac{\pi}{T_S} \quad \Rightarrow \quad T(\omega + \frac{2\pi i}{T_S}) = T(\omega)$$

$T(\omega)$ period: $2\pi/T_s$

Transfer function of transversal filter

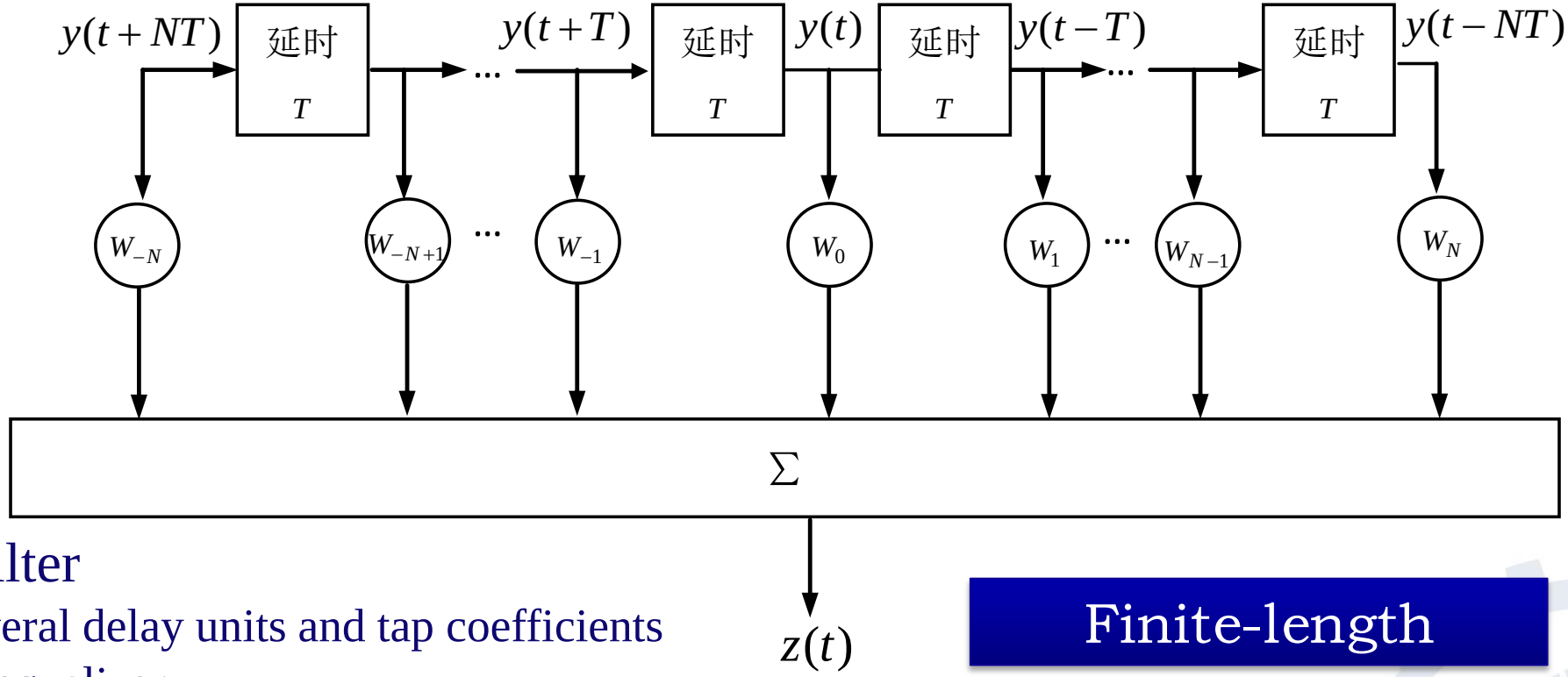
$$T(\omega) = \sum_{n=-\infty}^{\infty} W_n e^{-jnT_s\omega} \quad \Leftarrow \quad T(\omega) = \frac{T_S}{\sum_i H(\omega + \frac{2\pi i}{T_S})} \qquad |\omega| \le \frac{\pi}{T_s}$$

Impulse response of transversal filter

$$h_T(t) = \sum_{n=-\infty}^{\infty} W_n \delta(t - nT_S) \quad \Rightarrow \quad W_n = \frac{T_S}{2\pi} \int_{-\frac{\pi}{T_s}}^{\frac{\pi}{T_s}} \frac{T_s}{\sum_{n=-\infty}^{\infty} H(\omega - \frac{2n\pi}{T_s})} e^{jnT_s\omega} d\omega$$

Transversal Filter

$$h_T(t) = \sum_{n=-\infty}^{\infty} W_n \delta(t - nT_S)$$



- Transversal Filter
  - Consists of several delay units and tap coefficients
  - Tap delay line equalizer
- Tap delay: symbol period
- Weighted coefficient: adjustable

Finite-length

Tap Gain Adjustment of Transversal Filter

- Impulse response $$h_T(t) = \sum_{n=-N}^{N} W_n \delta(t - nT_S)$$
- Given input $y(t)$, supposing no additive noise, the output after equalization $z(t)$ is
$$z(t) = y(t) * h(t) = \sum_{i=-N}^{N} W_i y(t - iT_S)$$
- At sample moment $t = mT_s$ (assuming no delay in the system), then
$$z(mT_S) = \sum_{i=-N}^{N} W_i y(mT_S - iT_S) = \sum_{i=-N}^{N} W_i y[(m-i)T_S]$$
- According to 1st Nyquist criterion, then
$$z(mT_S) = \begin{cases} 1, m = 0 \\ 0, m = \pm 1, \pm 2, \ldots, \pm N \end{cases} \qquad \sum_{i=-N}^{N} W_i y[(m-i)T_S] = \begin{cases} 1, m = 0 \\ 0, m = \pm 1, \pm 2, \ldots, \pm N \end{cases}$$

Adjust tap gain based on W_i

Zero-forcing Equalization

$$\sum_{i=-N}^{N} W_i y[(m-i)T_S] = \begin{cases} 1, m = 0 \\ 0, m = \pm 1, \pm 2, ..., \pm N \end{cases} \Leftrightarrow \begin{bmatrix} y_0 & y_{-1} & \cdots & y_{-2N} \\ \vdots & \vdots & \cdots & \vdots \\ y_N & y_{N-1} & \cdots & y_{-N} \\ \vdots & \vdots & \vdots & \vdots \\ y_{2N} & y_{2N-1} & \cdots & y_0 \end{bmatrix} \begin{bmatrix} W_{-N} \\ W_{-N+1} \\ \vdots \\ W_0 \\ \vdots \\ W_{N-1} \\ W_N \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

- For given input sequence $\{y_k\}$, adjust $2N+1$ tap coefficients W_i
- So as to force N sample points y_k former and after equalization output z_0 have no ISI

Example

- Please design a 3-tap zero-forcing equalizer to inhibit ISI. Supposing $y_{-2} = 0$, $y_{-1} = 0.1$, $y_0 = 1$, $y_1 = -0.2$, $y_2 = 0.1$, please determine the coefficients.

2N+1=3,

$$\begin{bmatrix} y_0 & y_{-1} & y_{-2} \\ y_1 & y_0 & y_{-1} \\ y_2 & y_1 & y_0 \end{bmatrix} \begin{bmatrix} W_{-1} \\ W_0 \\ W_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{cases} W_{-1} + 0.1W_0 = 0 \\ -0.2W_{-1} + W_0 + 0.1W_1 = 1 \\ 0.1W_{-1} - 0.2W_0 + W_1 = 0 \end{cases}$$

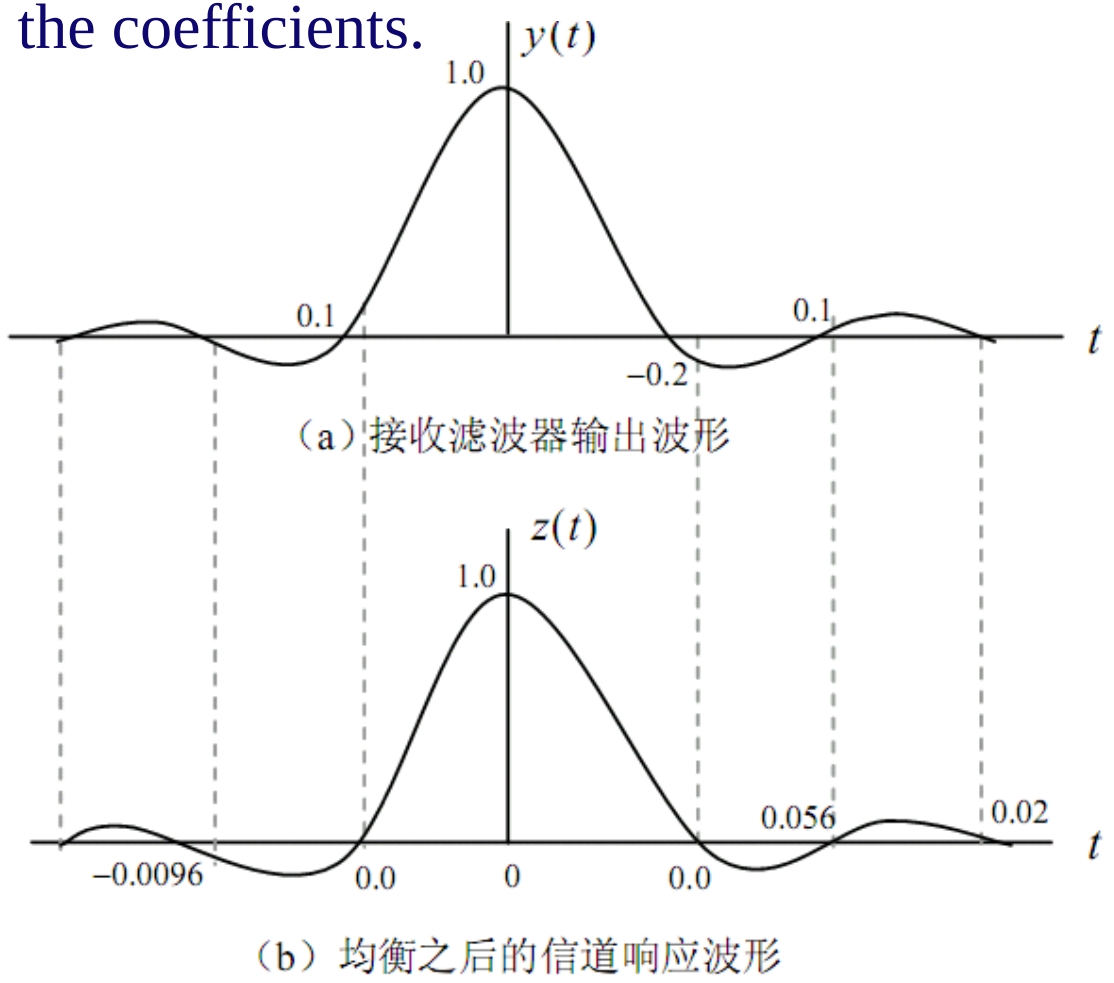
$$W_{-1} = -0.09606, \quad W_0 = 0.9606, \quad W_1 = 0.2017$$

$$z_k = \sum_{i=-N}^{N} W_i y_{k-i}$$

$z_{-1} = 0, \quad z_0 = 1, \quad z_1 = 0$

$z_{-3} = 0, \quad z_{-2} = 0.0096,$

$z_2 = 0.0557, \quad z_3 = 0.02016$



（a）接收滤波器输出波形

（b）均衡之后的信道响应波形

Summary

- Baseband Signal
  - Basic Code Types; Transmission Code Types; Power Spectrum
- Baseband System
  - ISI
  - 1st Nyquist Criterion
    - Ideal Lowpass System
    - Rolloff Lowpass System
  - 2nd Nyquist Criterion
    - Class I & IV Partial Response System
- Error Performance Analysis
  - Optimum Threshold Level
  - Symbol Error Rate
- Equalization
  - Transversal Filter

